Progress towards an open-source eigenvalue solver for free-boundary MHD stability

Maya Avida

Senior Thesis
Department of Physics
Princeton University
Spring 2025

Advised by Professor Egemen Kolemen

This paper represents my work in accordance with University regulations.

/s/ Maya Avida

Abstract

Analysis of MHD stability in stellarators is analytically and computationally complex; while analytic figures of merit can provide physical insight into the stability properties of an equilibrium, more precise stability computations are necessary for improved stellarator optimization. This work makes significant progress towards extending an MHD stability eigenvalue solver in the DESC stellarator optimization suite to consider stability to external modes. Numerical results from the MHD stability eigenvalue solver are then compared to analytic results for a simple test case, validating the eigenvalue solver and its extension to external modes.

Contents

I Magnetic confinement fusion **3**
- I.I Fusion energy . . . 3
- I.II Basic principles of magnetic confinement fusion . . . 3
- I.III Rotational transform . . . 6
- I.IV Plasma physics goals . . . 6

II Fluid models of plasmas **8**
- II.I Two-fluid model of a plasma . . . 8
- II.II Single-fluid magnetohydrodynamics . . . 9
- II.III Ideal MHD . . . 9
- II.IV Summary of assumptions in ideal MHD . . . 10
- II.V The flux freezing theorem . . . 10
- II.VI Ideal MHD equilibrium . . . 11

III Spectral and pseudospectral methods **12**

IV Stellarator optimization and DESC **14**
- IV.I Stellarators . . . 14
- IV.II DESC . . . 14

V Flux coordinate systems **16**
- V.I Flux coordinate systems . . . 16
- V.II Magnetic coordinate systems . . . 17

VI Linear MHD stability **20**
- VI.I The Energy Principle . . . 20
  - VI.I.i Derivation of the Energy Principle . . . 20
  - VI.I.ii Classification of instabilities . . . 23
- VI.II Stability in stellarators . . . 23
  - VI.II.i Overview . . . 23
  - VI.II.ii Numerical implementations of stability to internal modes . . . 24

VII Numerical implementations of MHD stability calculations in simple 1D models **26**
- VII.I Stability of the Z-pinch using the Energy Principle . . . 26
  - VII.I.i Z-pinch . . . 26
  - VII.I.ii Stability of the Z-pinch using the Energy Principle . . . 27

VII.I.iii Numerical implementation 28
VII.II Newcomb solver for a straight tokamak in DESC 29
VII.II.i Overview of Newcomb's procedure 30

VIII Implementation of external modes in the DESC stability solver 33
VIII.I Exterior Neumann problem 36
VIII.II Validation of solution to exterior Neumann problem 39
VIII.II.i Recreation of toy vacuum field in a non-axisymmetric geometry 39
VIII.II.ii Accurate integration of vacuum energy 40

IX Comparison to analytic results for a straight tokamak 43
IX.I Ordering 43
IX.II Analytic results and comparison to results from eigenvalue solver 43
IX.III Internal $m = 1$ kink mode 43
IX.III.i Analytic results 43
IX.III.ii Comparison to results from eigenvalue solver 44
IX.IV External kink modes 46
IX.IV.i Analytic results 46
IX.IV.ii Results from eigenvalue solver 48

X Conclusion 55

XI Future work 56

References 58

A Curvilinear coordinate systems 63

Acknowledgements

Thank you to my advisor, Professor Egemen Kolemen, for your mentorship and thoughtful guidance throughout this project, and for your immense contributions towards solving the hardest problems in fusion engineering across stellarator optimization, tokamak control, and liquid metals — your work is instrumental in putting fusion on the grid.

Thank you to Dr. Dario Panici and Yigit Elmacioglu for unwavering instruction throughout the process; this thesis would not have been possible without the many meetings spent overcoming obstacles in my path, and to Dario in particular for going above and beyond in providing help and support at all hours of the day and night.

Thank you to Dr. Rahul Gaur for help and guidance on obtaining, running, and understanding the AGNI code; I am incredibly grateful for the advice and support provided throughout the research process.

Thank you to Professor Robert Goldston, for igniting my interest in fusion through your course, and the support and advice you have provided ever since. Thank you to Dr. Daniel Dudt and Dr. David Gates, for the mentorship that kindled my interest in stellarators, inspiring me to dedicate my life to the subject.

Thank you to the Princeton Physics department, for nurturing my passion for physics, which has grown with every course I have taken, from Experimental Physics to Thermal Physics to Quantum Field Theory.

Thank you to my family: my mother, for being my role model for the past twenty-one years and for being my most consistent support system; my father, for making every moment special, whether we are snowboarding or at the DMV; and my sister, for being the person I know I can invariably count on when I really need it.

Thank you to the incredible friends I have made at Princeton. To name a small handful: thank you to Hanna, for inspiring me every day to be a kinder, more spontaneous, and warmer person — your goodness is infectiousness. Thank you to Brianna, for being the most incredible roommate, friend, and guardian angel I could ever ask for; I could not have survived these last four years without you. Thank you to Sidra, for countless fascinating conversations; we never run out of things to talk about. Thank you to Brian, for never failing to keep me entertained. Thank you to Colby, Connor, and Sam, for making even the latest nights spent studying fond memories (and to Connor in particular for proofreading this thesis). To the many other people here I have been lucky enough to know: thank you, and I look forward to continuing to remaining close well after graduation.

I Magnetic confinement fusion

I.I Fusion energy

It is imperative that the global energy system transition away from fossil fuels as quickly as possible, in order to minimize the adverse climate impacts of greenhouse gas emissions, as well as to ensure the security of the global energy supply. Controlled nuclear fusion presents an opportunity for a clean, reliable, and essentially limitless energy source. Tritium is produced locally from lithium-6, so that the fuels for D-T fusion — deuterium and lithium-6 — are found abundantly in nature and are not radioactive. Deuterium is a stable isotope of hydrogen and is found in all forms of water. Lithium-6 occurs in natural lithium with an abundance of 7.5%; there is enough ^{6}Li in seawater to produce 1.7×10^{8} TWy of fusion energy [1]; for context, the world consumes approximately 3 TWy of electricity annually [2]. Furthermore, the direct products of a fusion reaction — an α particle and a neutron — are also not radioactive; neutrons can activate the surrounding material, but the activated material is still much less radioactive than fission products. Fusion reactors have no risk of a meltdown, since any loss of confinement would quickly end the fusion reactions, precluding any possibility of an uncontrolled burn. A fusion power plant on the grid would therefore be a significant advancement for humankind, clearing the path towards a more sustainable future.

I.II Basic principles of magnetic confinement fusion

In order for fusion to occur, the deuterium and tritium nuclei must quantum mechanically tunnel through the Coulomb barrier, in order to be bound together by the shorter-range strong force. The nuclei therefore must be at extraordinarily high temperatures in order for the reaction to occur. At these temperatures, the deuterium and tritium fuels ionize and become plasma. Furthermore, in order to sustain the reaction and produce energy, the plasma must be confined. Because the charged particles in a fusion plasma are subject to electromagnetic forces, one very promising approach uses a toroidal magnetic field to confine a fusion plasma.

In a straight, uniform magnetic field, the charged particles in a plasma orbit the magnetic field lines with a cyclotron frequency of $\omega_c = \frac{|q||\mathbf{B}|}{m}$ and a Larmor radius of $r_L = \frac{|\mathbf{v}_\perp|}{\omega_c} = \frac{m|\mathbf{v}_\perp|}{|q||\mathbf{B}|}$ where the $\|$ and $\perp$ symbols reference the direction with respect to the magnetic field. In contrast, the motion along the magnetic field lines is unconstrained.

In a laboratory fusion plasma, the magnetic field is not straight and uniform; however, the length scale r_L is typically much smaller than the macroscopic length scale of the plasma L, and the cyclotron frequency is typically much larger than the characteristic frequency of motion along field lines $\frac{v_\parallel}{L}$, i.e. $r_L \ll L$, $\omega_c \gg \frac{v_\parallel}{L}$. Here, $v_\parallel \sim v_\perp \sim v_t = \sqrt{\frac{2T}{m}}$.

Then, to good approximation, the charged particles in a fusion plasma orbit rapidly around the field lines, while the guiding center of their orbital motion follows the magnetic field lines. If these field lines remain on closed surfaces, then the guiding center of the charged particles (to zeroth order in $\frac{r_L}{L}$) remain confined to these surfaces.

It is therefore advantageous for the magnetic field lines to lie on closed and nested surfaces, in order to confine the charged particles to these surfaces. These surfaces are known as *flux surfaces*. By the Poincare-Hopf theorem, the simplest closed surface embedded in 3D for which a vector field can lie everywhere tangent is a torus, so this is the most common topology chosen for magnetic confinement fusion (MCF) devices.

The *toroidal* direction is defined to be the long way around the torus, and the *poloidal* direction is defined to be the short way around the torus (i.e., a poloidal loop would link the torus, whereas a toroidal loop would not). See figure 1 for a diagram of the basic geometry and components of a *tokamak*, which is an *axisymmetric* toroidal MCF device (i.e. $\frac{\partial}{\partial \phi} = 0$, where ϕ is the toroidal angle in cylindrical coordinates).

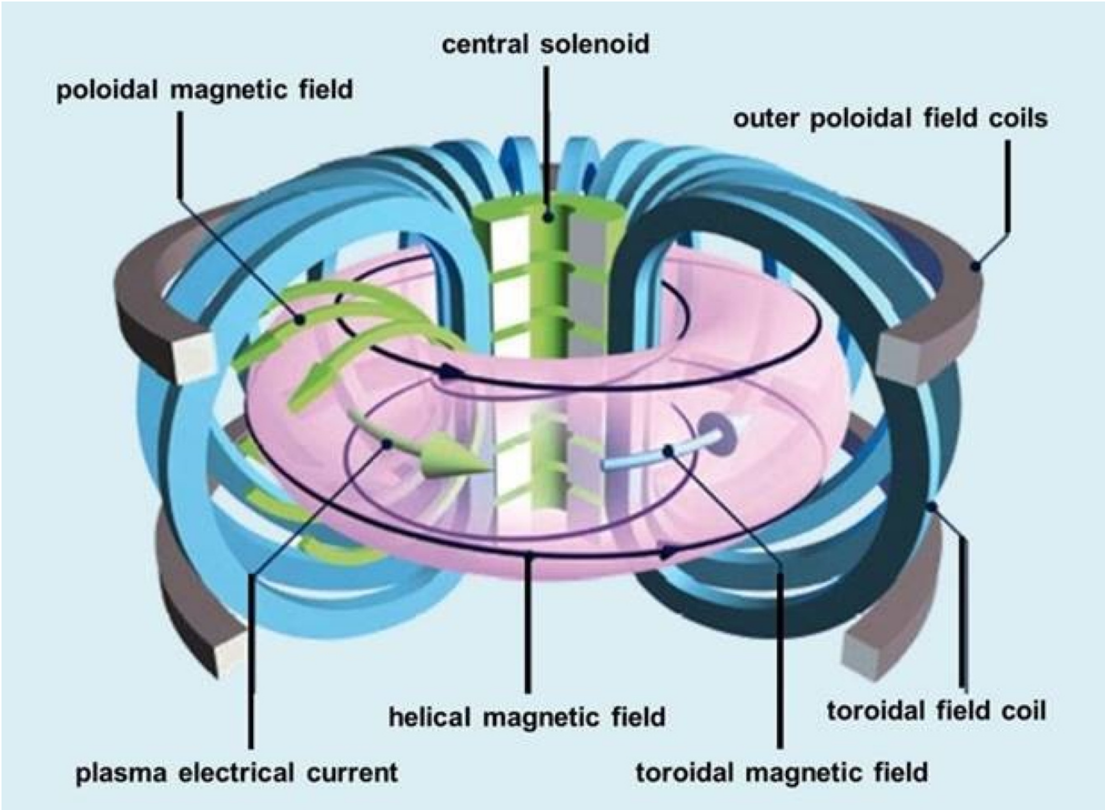

Figure 1: A diagram of a tokamak, an axisymmetric toroidal MCF device, from the U.S. Department of Energy [3]. The toroidal magnetic field is produced by toroidal field coils, whereas the poloidal magnetic field is generated primarily from the plasma current, which is driven by a central solenoid. These two field components sum to produce a helical magnetic field.

As discussed, to zeroth order in $\frac{r_L}{L}$, the guiding center trajectories of charged particles follow the magnetic field lines. However, to first order in $\frac{r_L}{L}$, the guiding center of these charged particles drift from the zeroth order trajectory along the field lines. The three first-order drifts are the $\mathbf{E}\times\mathbf{B}$ drift, the curvature drift, and the grad-B drift:

$$\mathbf{v}_{\mathbf{E}\times\mathbf{B}} = \frac{\mathbf{E}\times\hat{\mathbf{b}}}{B} \qquad \mathbf{v}_{\mathrm{curv}} = \pm\frac{v_\parallel^2}{\omega_c}\hat{\mathbf{b}}\times[(\hat{\mathbf{b}}\cdot\nabla)\hat{\mathbf{b}}] \qquad \mathbf{v}_{\mathrm{grad}} = \pm\frac{v_\perp^2}{2\omega_c}\hat{\mathbf{b}}\times\nabla\ln B$$

where $\pm$ indicates the sign of the particle charge. Observe that the grad-B and curvature drifts are dependent on particle charge and velocity, whereas the $\mathbf{E}\times\mathbf{B}$ drift is identical for all particles. If the device has a purely toroidal field, the curvature and grad-B drifts will then cause the ions and electrons to drift in opposite directions along the z-axis. This charge separation will then induce a vertically oriented electric field. The $\mathbf{E}\times\mathbf{B}$ drift then induces a radially outward drift of both the ions and electrons, and therefore a loss of confinement [4].

If, on the other hand, the field has a nonzero poloidal component, then each particle will alternate between the top and bottom of the device during its toroidal transits. Suppose a particle drifts upwards at the top of the device, onto an outer surface. Then — since the surfaces are nested — when the particle has followed its new magnetic field line to the bottom of the device, it will still drift upwards, returning to its original flux surface. Therefore, on average, the particle remains confined to a single surface. This illustrates the necessity of a nonzero poloidal magnetic field.

I.III Rotational transform

A key dimensionless parameter that quantifies the relationship between the poloidal and toroidal magnetic field is the *rotational transform*, which we will define now, as it will be used throughout this text. If we follow a given field line once toroidally, it will experience a change 2π in the toroidal angle (by definition) and a change $\Delta\theta$ in the poloidal angle. If we continue to follow the field line around the device, it will experience a change in the poloidal angle $(\Delta\theta)_k$ after the kth turn. The rotational transform then quantifies the ratio between the number of toroidal turns and the number of poloidal turns, and is therefore closely related to the ratio between the two fields [5]. Explicitly,

$$\iota = \lim_{n\to\infty} \frac{\sum_{k=1}^{n}(\Delta\theta)_k}{2\pi n} \tag{I.1}$$

As seen in I.II, a nonzero rotational transform is necessary for confinement.

I.IV Plasma physics goals

The primary goal of an MCF device is to produce energy. To this end, the primary physics figure of merit of a fusion device is $Q \equiv \frac{P_{\text{fusion}}}{P_{\text{external}}}$, i.e. the ratio of fusion power to external heating. To date, no MCF device has achieved $Q > 1$, i.e. scientific break-even. The goal of a fusion device would be a self-sustaining fusion reaction: $Q \to \infty$. This condition is known as ignition. A closely related figure of merit is the Lawson criterion $nT\tau_E$, where n is the number density ($n_i = n_e$ to maintain quasi-neutrality), T is the ion temperature, and $\tau_E \equiv \frac{U}{P_{\text{loss}}}$ is the energy confinement time. Here, U the stored energy in the plasma and P_{loss} is the rate of energy loss [6]. The condition for ignition is given by

$$nT\tau_E > 3 \times 10^{21} m^{-3}\text{keV s}$$

The Lawson criterion therefore underscores the importance in maintaining good particle confinement, particularly of the energetic α-particles produced during the fusion reaction, and overall minimizing particle transport. Furthermore, in order to sustain the reaction for as long as possible, it is necessary that the device not just operate in an equilibrium, but that this force balance equilibrium be stable to perturbations to the plasma volume. It is this latter goal that will be the focus of this paper.

II Fluid models of plasmas

As mentioned above, at fusion-relevant temperatures, the deuterium and tritium gas in a fusion device have ionized and become plasma. Each particle in the plasma interacts with every other particle in the plasma through long and short-range Coulomb interactions, producing both an electric field and a magnetic field. Given the vast quantity of particles in a fusion device, modeling each particle individually is of course computationally intractable.

It is advantageous to develop simplified models of plasma behavior. One comparatively simple model of a plasma treats it as an electrically conducting fluid, producing a set of magnetohydrodynamic equations. This section generally follows Goldston and Rutherford [4].

II.I Two-fluid model of a plasma

In the following subsection, for a given fluid species α (typically ions and electrons in a fully ionized plasma), we will define

1. The number density n_α
2. The mean velocity at a point $\mathbf{u}_\alpha$
3. The magnetic field $\mathbf{B}$
4. The electric field $\mathbf{E}$
5. The mass of each particle m_α
6. The pressure tensor $\overleftrightarrow{\mathbf{P}}_\alpha$, which will generally be treated as a scalar p_α
7. The volumetric plasma current density $\mathbf{J}$
8. The rate of momentum exchange between α and another particle species β $\mathbf{R}_{\alpha\beta}$. In the case of a two-species plasma with ions and electrons, $\mathbf{R}_{ei} = \eta n_e e \mathbf{J}$ where η is the resistivity.

We will begin by explaining the most general set of fluid equations for a plasma. Continuity requires

$$\frac{\partial n_\alpha}{\partial t} + \nabla \cdot (n_\alpha \mathbf{u}_\alpha) = S_\alpha$$

Where S_α is a source or sink term arising from ionization or recombination, that will be neglected from this point forward, i.e. $S_\alpha = 0$.

The Lorentz force and pressure forces require that the momentum density evolve as

$$\frac{\partial(m_\alpha n_\alpha \mathbf{u}_\alpha)}{\partial t} + \nabla\cdot(m_\alpha n_\alpha \mathbf{u}_\alpha \mathbf{u}_\alpha) = n_\alpha q_\alpha(\mathbf{E} + \mathbf{u}_\alpha \times \mathbf{B}) - \nabla\cdot\overleftrightarrow{\mathbf{P}}_\alpha + \sum_\beta \mathbf{R}_{\alpha\beta}$$

In the case of isotropic pressure, $\nabla\cdot\overleftrightarrow{\mathbf{P}}_\alpha = \nabla p_\alpha$. The pressure p_α is generally approximated by the equation of state $p_\alpha = Cn_\alpha^\gamma$, which describes how the plasma temperature responds to compression. γ can take on a variety of values depending on the timescale of compression. In the case of adiabatic compression, $\gamma = \frac{5}{3}$.

II.II Single-fluid magnetohydrodynamics

Linear combinations of the equations in II.I, under the assumption that the ion Larmor radius is very small compared to the length scale of fluid motion and ignoring electron inertia entirely, give the following equations:

$$\mathbf{E} + \mathbf{u}\times\mathbf{B} = \eta\mathbf{J} \tag{II.1a}$$

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{u}) = 0 \tag{II.1b}$$

$$\rho\frac{D\mathbf{u}}{dt} = \mathbf{J}\times\mathbf{B} - \nabla p \tag{II.1c}$$

$$\nabla\times\mathbf{B} = \mu_0\mathbf{J} \tag{II.1d}$$

$$\nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} \tag{II.1e}$$

$$\nabla\cdot\mathbf{B} = 0 \tag{II.1f}$$

$$p = Cn^\gamma \tag{II.1g}$$

Here, $p = \sum_\alpha p_\alpha$ is the pressure, $\rho = \sum_\alpha m_\alpha n_\alpha$ is the mass density, $\mathbf{u} = \frac{1}{\rho}\sum_\alpha \rho_\alpha \mathbf{u}_\alpha$ is the center-of-mass velocity and $\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + \mathbf{u}\cdot\nabla$ is the convective derivative. We have used the non-relativistic limit of Maxwell's equations, i.e. dropped the displacement current $\varepsilon_0\frac{\partial\mathbf{E}}{\partial t}$.

II.III Ideal MHD

In the limit of infinite conductivity, i.e. $\eta \to 0$, we have $\mathbf{E} + \mathbf{u}\times\mathbf{B} = 0$. This limit is equivalent to the statement that $L \gg \lambda_\eta$, where λ_η is the length scale where resistive effects become important, i.e. where $\mathbf{B}\cdot\nabla\mathbf{u} \sim \eta\nabla^2\mathbf{B}$.

This limit is known as *ideal magnetohydrodynamics*, and the assumption of infinite conductivity will be employed for the remainder of this paper.

II.IV Summary of assumptions in ideal MHD

To summarize, here are the assumptions we have made in the derivation of ideal MHD [7] [8].

Quasineutrality The plasma is very nearly charge-neutral; $\sum_\alpha q_\alpha n_\alpha \sim 0$, and all relevant length scales are much larger than the Debye length $\lambda_D = \sqrt{\frac{\varepsilon_0 T_e}{n_e e^2}}$.

Non-relativistic The displacement current has been dropped.

Infinite conductivity We are considering length scales significantly larger than the scales at which resistive dissipation become relevant.

Ions dominate mass and momentum density When taking linear combinations of the single-fluid equations, we have ignored the nonlinear drift term $\nabla \cdot (\sum_\alpha m_\alpha n_\alpha (\mathbf{u} - \mathbf{u}_\alpha)(\mathbf{u} - \mathbf{u}_\alpha))$ by assuming that the electron mass is sufficiently small to be negligible.

Small Larmor radius This assumption allows us to drop the Hall field, the thermoelectric field, and the electron inertia term in the expression for the electric field II.1a.

High collisionality The most concerning assumption is that of a short mean free path: $\lambda_{\mathrm{mfp}} \ll L$. This is the underlying assumption that makes fluid models useful, since neighboring particles interact sufficiently frequently that they can be treated as a single fluid element, allowing the plasma to be treated as locally Maxwellian with isotropic pressure. This condition is never satisfied in fusion plasmas — however, we will discover that most instabilities in fusion plasmas are incompressible ($\nabla \cdot \mathbf{u} = 0$). Under these conditions, the collisional and collisionless equations of state are identical.

II.V The flux freezing theorem

One of the key properties of the ideal MHD equations is that the topology of the magnetic field is fixed and moves with the plasma. Define the flux through an open surface S_p bounded by an arc $\mathbf{l}$ as

$$\psi = \int_{S_p} \mathbf{B} \cdot \mathbf{n} dS$$

Now suppose that the plasma on the surface S_p is moving with velocity $\mathbf{v}$. Then

$$\frac{d\psi}{dt} = \int_{S_p} \frac{\partial \mathbf{B}}{\partial t} \cdot \mathbf{n} dS - \oint \mathbf{v} \times \mathbf{B} \cdot d\mathbf{l}$$

From Faraday's law, $\int_{S_p} \frac{\partial \mathbf{B}}{\partial t} \cdot \mathbf{n} dS = -\int_{S_p} (\nabla \times \mathbf{E}) \cdot \mathbf{n} dS$ so

$$\frac{d\psi}{dt} = \oint (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l} = 0 \tag{II.2}$$

From (II.1a) with $\eta = 0$.

This derivation applies to any surface, so that the magnetic flux through any cross-section of a long, thin flux tube is conserved. It then follows that the magnetic field lines move with the plasma in ideal MHD, and their topology is fixed.

II.VI Ideal MHD equilibrium

By imposing time-independence and $\mathbf{u} = 0$ on the ideal MHD equations, we obtain

$$\mathbf{J} \times \mathbf{B} = \nabla p \tag{II.3a}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \tag{II.3b}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{II.3c}$$

This is the set of partial differential equations for ideal MHD equilibrium.

III Spectral and pseudospectral methods

Spectral and pseudospectral methods are a family of techniques used for solving differential and integral equations by approximating the solution as a finite linear combination of global basis functions, such as a truncated Fourier series or series of Chebyshev polynomials (defined by $T_n(\cos\theta) = \cos(n\theta)$). Derivatives are then computed by taking analytic derivatives of the spectral fit.

Pseudospectral methods are a type of spectral method, in which the known values of a function at a grid of N points are related to the coefficients of N basis functions. At these N grid points, the spectral decomposition matches the known values of the function exactly. The grid points are known as *collocation nodes.*

With a sufficiently smooth function, a well-chosen spectral basis, and a well-chosen set of collocation nodes, pseudospectral methods have the property of infinite-order or exponential convergence, i.e.

$$\text{Pseudospectral error} \sim O\left(\frac{1}{N^N}\right)$$

More precisely, Darboux's principle states that both the domain of convergence in the complex plane and the rate of convergence in the complex plane are controlled by the singularities of $f(x)$ in the complex plane. In general, a spectral series will converge to $f(x)$ in the largest domain of the complex plane possible, with the shape fixed by the basis set [9]. For a standard Taylor series, the shape of the convergence domain in the complex plane is a disk; therefore, even if the function is smooth everywhere along the real axis, if $f(x)$ has a singularity at $x = i$ (for instance), the Taylor series of $f(x)$ will not converge for $|x| > 1$. In contrast, the shape of the convergence domain for a Fourier series is an infinite strip on the real axis. This implies that if the spatially periodic function $f(x)$ is infinitely differentiable $\forall\ x \in \mathbb{R}$, then the Fourier series will have infinite-order convergence to f in that domain.

Likewise, for a series of Chebyshev polynomials, the convergence domain is an ellipse with foci at $x = \pm 1$. Therefore, a series of Chebyshev polynomials will always converge for all $x \in [-1, 1]$, as long as $f(x)$ is sufficiently smooth in this domain. Specifically, if all derivatives of f are bounded on $x \in [-1, 1]$, then a Chebyshev series is also guaranteed infinite-order convergence.

Because the pseudospectral error decreases so quickly as a function of the number of grid points, pseudospectral methods tend to require significantly less memory than finite-difference methods. This property will be critical for the eigenvalue solver described later on in VI.II.ii, as the matrices required are exceptionally memory-intensive and the memory requirements are usually the limiting factor on the spatial resolution of the solution.

IV Stellarator optimization and DESC

IV.I Stellarators

As described in I.II, $\iota > 0$ is necessary for particle confinement. In a tokamak, which is axisymmetric, the only way to generate rotational transform is through a driven plasma current. Typically, this requires the use of a transformer to drive a current in the plasma, inherently forcing the device to operate in pulsed, rather than steady-state, operation (Figure 1). Additionally, as we will see later, plasma current is a significant driver of instabilities, so that tokamaks are prone to catastrophic MHD instabilities known as disruptions. On the other hand, one significant advantage of axisymmetry is that it leads to a conserved momentum in the guiding center Lagrangian of a particle, guaranteeing particle confinement. Axisymmetry also guarantees the existence of nested flux surfaces [5].

In contrast, a *stellarator* is not axisymmetric, so that rotational transform can be produced through torsion of the magnetic axis or rotation of magnetic surfaces [5]. Because they do not require a driven plasma current, stellarators intrinsically operate in steady-state and have much more favorable stability properties. Even when a stability criterion is violated, the device typically does not disrupt; rather, transport increases due to MHD turbulence, degrading performance but posing a significantly smaller risk of damage to the material components of the device [7]. On the other hand, due to their lack of axisymmetry, stellarators are not guaranteed favorable particle confinement or nested flux surfaces. For this reason, until the advent of modern numerical optimization techniques, designing effective stellarators was incredibly difficult. However, through the use of optimization, stellarators' large parameter space can be explored in order to design an effective fusion device. Through optimizing for hidden symmetries such as quasisymmetry [10], confinement of particle guiding centers can also be obtained without relying on axisymmetry.

Tokamaks are frequently referred to as two-dimensional devices, due to the symmetry in the toroidal direction. In contrast, stellarators are frequently referred to as three-dimensional.

IV.II DESC

DESC is a suite of codes written in Python used for stellarator optimization [11]. Rather than using a variational principle, DESC solves the force balance equation (II.3a) locally using pseudospectral methods, enabling fast and accurate equilibrium calculations.

We will see in V.II that the magnetic field can be expressed in terms of the geometry of the flux surfaces and the stream function λ. The current density can then be computed from Ampere's law (II.3b) using (A.10). The geometry of the flux surfaces $R(\rho, \theta, \zeta), Z(\rho, \theta, \zeta), \lambda(\rho, \theta, \zeta)$ is described using a spectral fit in terms of the DESC coordinates (ρ, θ, ζ). The spectral coefficients for R, Z, and λ are then optimized to minimize the force balance error $\mathbf{J} \times \mathbf{B} - \nabla p$ locally throughout the equilibrium, in order to satisfy (II.3a). The geometry of the flux surfaces is represented by a combined Fourier-Zernike basis, using a standard Fourier decomposition in the toroidal direction, and Zernike polynomials in the radial-poloidal plane [12]. This is a convenient basis set for a toroidal topology, as Zernike polynomials automatically satisfy regularity conditions at the magnetic axis.

The stellarator equilibrium is determined by the geometry of the plasma boundary, the pressure profile, and either the rotational transform or toroidal current profile [13]. In DESC, the user can optimize the equilibrium with respect to these inputs for stability figures of merit, fast particle confinement, and engineering complexity. Leveraging automatic differentiation in JAX [14], DESC can accurately and efficiently compute gradients of complex objectives, enabling fast and accurate equilibrium calculations and optimization.

V Flux coordinate systems

Throughout this section, results and definitions from Appendix A will be referenced without explicit explanation.

V.I Flux coordinate systems

A fundamental assumption in DESC, and much stellarator theory, is that closed and nested toroidal flux surfaces exist. The center of these flux surfaces is a one-dimensional curve known as the *magnetic axis*. Under this assumption, we can adopt a new and convenient coordinate system, called a *flux coordinate system*. In these coordinates, the first coordinate is a radius-like coordinate that labels each flux surface. The other two coordinates describe the position on each flux surface, most commonly a poloidal angle and a toroidal angle (Figure 2).

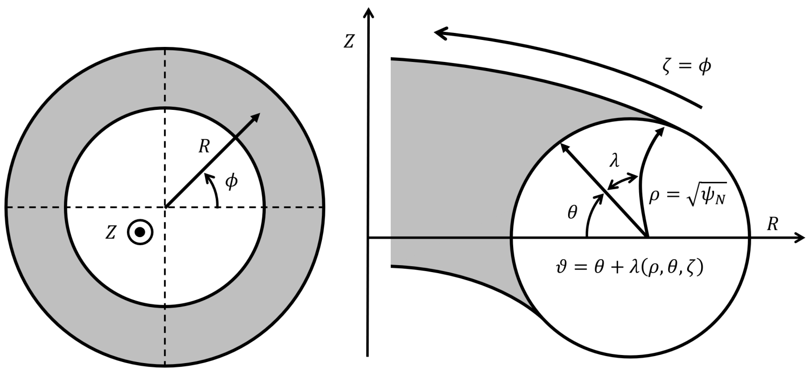

Figure 2: Coordinate system used by the DESC code [12].

In DESC, the particular choice of flux coordinates is (ρ, θ, ζ), where ρ is the flux surface label, θ is a generalized poloidal angle, and $\zeta = \phi$ is the toroidal angle in cylindrical coordinates. Because $\mathbf{B}$ lies tangent to each flux surface, this coordinate system satisfies $\mathbf{B} \cdot \nabla\rho = 0$. A quantity that depends only on ρ is known as a *flux function*. For now, ρ is just a label, but it will later be explicitly defined in terms of the normalized toroidal flux ψ:

$$\psi(\rho) = \frac{1}{2\pi} \int_{S_T} d\mathbf{S} \cdot \mathbf{B} = \frac{1}{(2\pi)^2} \int_{\mathcal{V}} d\mathbf{r}\mathbf{B} \cdot \nabla\zeta$$

Here, S_T is a surface at constant ζ, bounded by the flux surface ρ, and $\mathcal{V}$ is the volume bounded by the flux surface ρ. Likewise,

$$\chi(\rho) = \frac{1}{2\pi}\int_{S_P} d\mathbf{S}\cdot\mathbf{B} = \frac{1}{(2\pi)^2}\int_{\mathcal{V}} d\mathbf{r}\mathbf{B}\cdot\nabla\theta$$

is the normalized enclosed poloidal flux. Here, S_P is a surface at constant θ, bounded by the flux surface ρ. The fact that ψ and χ are functions only of ρ follows from the condition $\nabla\cdot\mathbf{B} = 0$, Gauss' theorem, and $\mathbf{B}\cdot\nabla\rho = 0$.
Finally, ρ can be defined explicitly by

$$\rho \equiv \sqrt{\frac{\psi(\rho)}{\psi_{\text{total}}}} \equiv \sqrt{s} \in [0,1]$$

This particular choice of ρ is convenient because it is proportional to the minor radius r for a device with circular cross-section. Note that $\rho = 0$ at the magnetic axis, and $\rho = 1$ at the outermost flux surface.

We can also redefine the rotational transform ι by

$$\iota(\rho) = \frac{d\chi}{d\psi} \tag{V.1}$$

The equivalence of this definition to (I.1) will be demonstrated in the following subsubsection.

V.II Magnetic coordinate systems

Using $\mathbf{B}\cdot\nabla\rho = 0$, i.e. $\mathbf{B} = B^\theta\mathbf{e}_\theta + B^\zeta\mathbf{e}_\zeta$ and (A.9), the divergence-free condition becomes

$$0 = \nabla\cdot\mathbf{B} = \frac{1}{\sqrt{g}}\left(\frac{\partial(\sqrt{g}B^\theta)}{\partial\theta} + \frac{\partial(\sqrt{g}B^\zeta)}{\partial\zeta}\right)$$

with $\mathbf{B}$ doubly periodic in θ and ζ. A divergence-free vector field in two dimensions can be expressed in terms of a stream function:

$$\sqrt{g}\mathbf{B} = -\frac{\partial\alpha}{\partial\zeta}e_\theta + \frac{\partial\alpha}{\partial\theta}e_\zeta \Longleftrightarrow \mathbf{B} = e^\rho\times\nabla\alpha \tag{V.2}$$

Since $\mathbf{B}$ is doubly periodic, α must have secular terms in θ and ζ and a doubly periodic term λ. Specifically,

$$\alpha(\rho,\theta,\zeta) = j(\rho)\theta + h(\rho)\zeta + \lambda(\rho,\theta,\zeta) \tag{V.3}$$

Then, using the periodicity of λ, we can relate the flux function j to the normalized toroidal flux:

$$
\begin{aligned}
\psi(\rho) &= \frac{1}{(2\pi)^2}\int_{\mathcal{V}} d\mathbf{r}\mathbf{B}\cdot\nabla\zeta \\
&= \frac{1}{2\pi}\int_0^\rho\int_0^{2\pi}\sqrt{g}B^\zeta d\theta d\rho' \\
&= \frac{1}{2\pi}\int_0^\rho\int_0^{2\pi}\frac{\partial\alpha}{\partial\theta}d\theta d\rho' \\
&= \int_0^\rho j(\rho)d\rho
\end{aligned}
$$

so $j(\rho) = \frac{d\psi}{d\rho}$. Similarly,

$$
\begin{aligned}
\chi(\rho) &= \frac{1}{(2\pi)^2}\int_{\mathcal{V}} d\mathbf{r}\mathbf{B}\cdot\nabla\theta \\
&= \frac{1}{2\pi}\int_0^\rho\int_0^{2\pi} d\zeta d\rho'\sqrt{g}B^\theta \\
&= -\frac{1}{2\pi}\int_0^\rho\int_0^{2\pi} d\zeta d\rho'\frac{\partial\alpha}{\partial\zeta} \\
&= -\int_0^\rho h(\rho)d\rho
\end{aligned}
$$

So $h(\rho) = -\frac{d\chi}{d\rho} = -\iota\frac{d\psi}{d\rho}$. Then from (V.2) and (V.3), we have

$$
\begin{aligned}
\mathbf{B} &= \mathbf{e}^\rho\times\left(\frac{d\psi}{d\rho}\mathbf{e}^\theta - \iota\frac{d\psi}{d\rho}\mathbf{e}^\zeta + \nabla\lambda\right) \\
&= \nabla\psi\times(\nabla\theta - \iota\nabla\zeta + \nabla\lambda)
\end{aligned}
$$

If we then define $\vartheta \equiv \theta + \lambda$, we have

$$
\begin{aligned}
\mathbf{B} &= \nabla\psi\times(\nabla\vartheta - \iota\nabla\zeta) && \text{(V.4a)} \\
&= \nabla\psi\times\vartheta + \nabla\zeta\times\nabla\chi && \text{(V.4b)} \\
&= \nabla\psi\times\nabla\vartheta + \nabla\zeta\times\nabla\chi && \text{(V.4c)} \\
&= \psi'(\rho)\left(\nabla\rho\times\nabla\vartheta + \iota\nabla\zeta\times\nabla\rho\right) && \text{(V.4d)} \\
&= \frac{\psi'(\rho)}{\sqrt{g}_m}\left(\mathbf{e}_\zeta + \iota\mathbf{e}_\vartheta\right) && \text{(V.4e)}
\end{aligned}
$$

Where $\sqrt{g}_m$ is the Jacobian of the (ρ, ϑ, ζ) coordinate system. (V.4e) then implies $\frac{B^\vartheta}{B^\zeta} = \iota(\rho)$, which is constant on a flux surface.

Now, if we parameterize a field line l by $\mathbf{R}(\psi_l, \vartheta_l(\zeta), \zeta)$, then we must have by the chain rule

$$\frac{d\mathbf{R}}{d\zeta} = \frac{d\vartheta_l}{d\zeta}\mathbf{e}_\vartheta + \mathbf{e}_\zeta$$

By definition, we are integrating a field line around the flux surface, so $\mathbf{B} \propto \frac{d\mathbf{R}}{d\zeta}$. Then

$$\frac{\psi'(\rho)}{\sqrt{g}_m}(\iota\mathbf{e}_\vartheta + \mathbf{e}_\zeta) = \mathbf{B} \propto \frac{d\mathbf{R}}{d\zeta} = \frac{d\vartheta_l}{d\zeta}\mathbf{e}_\vartheta + \mathbf{e}_\zeta$$

So that

$$\frac{d\vartheta}{d\zeta} = \iota \tag{V.5}$$

Therefore, in (ρ, ϑ, ζ) coordinates, **magnetic field lines are straight with slope $\iota(\rho)$ on any given flux surface**. Flux coordinate systems with this property are known as *magnetic coordinates*. Observe also that the equivalence of the two definitions of ι — (I.1) and (V.1) — is now clear.

The numerical implementation of MHD stability will take place in magnetic coordinates. The specific magnetic coordinate system used in this implementation is the PEST coordinate system (ρ, ϑ, ζ), which uses $\zeta = \phi$, i.e. the toroidal angle in cylindrical coordinates. The simple form for $\mathbf{B}$ given by (V.4e) will be very convenient for the numerical implementation of MHD stability.

Note from (V.5) that when $\iota = \frac{n}{m} \in \mathbb{Q}$, the field lines close on themselves after n toroidal transits, whereas when $\iota \in \mathbb{R}\backslash\mathbb{Q}$, the field line will fill the surface. Field lines that close on themselves are referred to as *rational field lines*, and surfaces with $\iota \in \mathbb{Q}$ are called *rational surfaces* or *resonant surfaces.*

VI Linear MHD stability

VI.I The Energy Principle

VI.I.i Derivation of the Energy Principle

Suppose we have found an equilibrium we like, and we would like to find out if it is stable to perturbations. If we linearize all quantities as

$$Q(\mathbf{r},t) = Q_0(\mathbf{r}) + \tilde{Q}_1(\mathbf{r},t) = Q_0(\mathbf{r}) + Q_1(\mathbf{r})e^{-i\omega t} \tag{VI.1}$$

Where $\tilde{Q}_1(\mathbf{r},t)$ is a small first-order perturbation, and ω is an eigenvalue that we would like to determine. The equilibrium is unstable if $\text{Im}(\omega) > 0$, i.e. the perturbation grows with time. We begin from the ideal MHD equilibrium conditions (II.3) and linearize all quantities about the equilibrium. We express all perturbed quantities in terms of the perturbation vector $\tilde{\boldsymbol{\xi}}(\mathbf{r},t)$, with

$$\mathbf{v}_1 = \frac{\partial\tilde{\boldsymbol{\xi}}}{\partial t}$$

Where $\tilde{\boldsymbol{\xi}}(\mathbf{r},0) = 0$ but $\frac{\partial\tilde{\boldsymbol{\xi}}(\mathbf{r},0)}{\partial t} \neq 0$. Here, $\tilde{\boldsymbol{\xi}}$ represents the displacement of the plasma from its equilibrium position. The perturbed, linearized MHD equations become

$$\rho\frac{\partial^2\tilde{\boldsymbol{\xi}}}{\partial t^2} = \mathbf{F}(\tilde{\boldsymbol{\xi}}) \equiv \mathbf{J}_0 \times \tilde{\mathbf{B}}_1 + \tilde{\mathbf{J}}_1 \times \mathbf{B}_0 - \nabla\tilde{p}_1 \tag{VI.2a}$$

$$\tilde{\rho}_1 = -\nabla\cdot(\rho_0\tilde{\boldsymbol{\xi}}) \tag{VI.2b}$$

$$\tilde{p}_1 = -\tilde{\boldsymbol{\xi}}\cdot\nabla p_0 - \gamma p_0\nabla\cdot\tilde{\boldsymbol{\xi}} \tag{VI.2c}$$

$$\tilde{\mathbf{B}}_1 = \nabla\times(\tilde{\boldsymbol{\xi}}\times\mathbf{B}_0) \tag{VI.2d}$$

Which, in the eigenvalue formulation (VI.1), becomes

$$-\omega^2\rho\boldsymbol{\xi} = \mathbf{F}(\boldsymbol{\xi}) = \frac{1}{\mu_0}(\nabla\times\mathbf{B})\times\mathbf{B}_1 + \frac{1}{\mu_0}(\nabla\times\mathbf{B}_1)\times\mathbf{B} + \nabla(\boldsymbol{\xi}\cdot\nabla p_0 + \gamma p_0\nabla\cdot\boldsymbol{\xi}) \tag{VI.3a}$$

$$\mathbf{B}_1 = \nabla\times\boldsymbol{\xi}\times\mathbf{B}_0 \tag{VI.3b}$$

It is a critical property that $\mathbf{F}(\boldsymbol{\xi})$ is a self-adjoint operator. This property has the following consequences:

1. The eigenvalues ω^2 of $\mathbf{F}(\boldsymbol{\xi})$ are purely real. Therefore, the system is unstable when $\omega^2 < 0$.

2. The eigenvectors $\boldsymbol{\xi}$ are orthogonal, so that any initial perturbation will always eventually be dominated by the fastest growing mode.

The spectrum of $\mathbf{F}(\boldsymbol{\xi})$ consists of both discrete and continuous modes. However, the continua occur only when $\omega^2 \geq 0$ and are therefore irrelevant to instabilities. If we define

$$\delta W(\boldsymbol{\xi}^*, \boldsymbol{\xi}) = -\frac{1}{2} \int \boldsymbol{\xi}^* \cdot \mathbf{F}(\boldsymbol{\xi}) d\mathbf{r} \tag{VI.4}$$

$$K = \frac{1}{2} \int \rho |\boldsymbol{\xi}|^2 d\mathbf{r} \tag{VI.5}$$

Then any trial function $\boldsymbol{\xi}$ (whether it is an eigenfunction or not) produces an estimate of the eigenvalue

$$\omega^2 = \frac{\delta W(\boldsymbol{\xi}^*, \boldsymbol{\xi})}{K(\boldsymbol{\xi}^*, \boldsymbol{\xi})} \tag{VI.6}$$

The variational method states that any allowable trial function that produces an extremum in ω^2 is itself an eigenfunction of the normal mode problem (VI.1) with eigenvalue ω^2. This can be used to prove the Energy Principle, which states that an equilibrium is stable if and only if

$$\delta W(\boldsymbol{\xi}^*, \boldsymbol{\xi}) \geq 0$$

For all possible perturbations $\boldsymbol{\xi}$, where

$$\delta W(\boldsymbol{\xi}^*, \boldsymbol{\xi}) = \delta W_F + \delta W_S + \delta W_V$$

is the total fluid, surface, and vacuum energy, given by

$$\delta W_F(\boldsymbol{\xi}^*, \boldsymbol{\xi}) \qquad \text{fluid energy} \tag{VI.7a}$$

$$= \frac{1}{2\mu_0} \int_P \Big[|\mathbf{Q}_\perp|^2 \qquad \text{shear Alfven wave} \tag{VI.7b}$$

$$+ B^2 |\nabla \cdot \boldsymbol{\xi}_\perp + 2\boldsymbol{\xi}_\perp \cdot \kappa|^2 \qquad \text{compressional Alfven wave} \tag{VI.7c}$$

$$+ \mu_0 \gamma p |\nabla \cdot \boldsymbol{\xi}|^2 \qquad \text{sound wave} \tag{VI.7d}$$

$$- 2\mu_0 (\boldsymbol{\xi}_\perp \cdot \nabla p)(\boldsymbol{\xi}_\perp^* \cdot \boldsymbol{\kappa}) \qquad \text{pressure-driven modes} \tag{VI.7e}$$

$$- \mu_0 J_\parallel (\boldsymbol{\xi}_\perp^* \times \mathbf{b}) \cdot \mathbf{Q}_\perp(\boldsymbol{\xi}_\perp) \Big] d\mathbf{r} \qquad \text{current-driven modes} \tag{VI.7f}$$

$$\delta W_S(\boldsymbol{\xi}^*, \boldsymbol{\xi}) \tag{VI.7g}$$

$$= \frac{1}{2\mu_0} \int_{S_p} |\mathbf{n} \cdot \boldsymbol{\xi}_\perp|^2 \mathbf{n} \cdot \left[\left[\nabla \left(\frac{B^2}{2} + \mu_0 p \right) \right]\right] dS \qquad \text{surface sheet current} \tag{VI.7h}$$

$$\delta W_V(\boldsymbol{\xi}^*, \boldsymbol{\xi}) \tag{VI.7i}$$

$$= \frac{1}{2\mu_0} \int_V |\hat{\mathbf{B}}_1|^2 d\mathbf{r} \qquad \text{vacuum magnetic energy} \tag{VI.7j}$$

The perturbation to the magnetic field can be given by

$$\mathbf{Q}(\boldsymbol{\xi}_\perp) = \nabla \times (\boldsymbol{\xi}_\perp \times \mathbf{B}) \quad \text{for } \mathbf{Q}\text{, the perturbed magnetic field inside the plasma} \tag{VI.8}$$

$$\nabla \cdot \hat{\mathbf{B}}_1 = \nabla \times \hat{\mathbf{B}}_1 = 0 \quad \text{for } \hat{\mathbf{B}}_1\text{, the perturbed magnetic field outside the plasma} \tag{VI.9}$$

Stabilizing terms in δW_F The first three terms in δW_F are all nonnegative, i.e. stabilizing. (VI.7b) corresponds to the restoring force from magnetic tension — field line bending is energetically intensive. In a uniform plasma, this term creates the shear Alfven wave, which describes a pure oscillation between the perturbed kinetic energy and the magnetic tension. (VI.7c) describes the energy required to compress magnetic field lines. Finally (VI.7d) corresponds to the energy required to compress the plasma itself; in general, the least stable mode in tokamaks and stellarators is incompressible ($\nabla \cdot \boldsymbol{\xi} = 0$), so this term typically vanishes.

The final two terms can be either positive or negative, and therefore are the driving terms of MHD instability. Specifically, when the pressure gradient aligns with the curvature $\boldsymbol{\kappa}$, for instance in the outboard side of a tokamak, the term (VI.7e) can be destabilizing. Similarly, we can see in (VI.7f) that current parallel to the magnetic field is also destabilizing.

A few other notes [7]:

1. The minimizing perturbation satisfies $\mathbf{B}\cdot\nabla(\nabla\cdot\boldsymbol{\xi}) = 0$. In any configuration in which the field lines lie on surfaces, but are only closed on isolated surfaces, this additionally implies $\nabla\cdot\boldsymbol{\xi} = 0$. Therefore, in most fusion configurations of interest, we can treat the perturbation as incompressible. The adiabatic compressibility of the plasma (a consequence of the invalid assumption of high collisionality) is therefore unimportant in the most unstable modes. Therefore, we can conjecture that incompressible MHD stability predictions are accurate in our configurations of interest.

2. The vacuum region assumption is more conservative than the true case of a resistive scrape-off layer, in which the marginal stability boundary is identical but the instability growth rates are smaller.

VI.I.ii Classification of instabilities

Instabilities are classified in a few ways. First, an instability can be an *external* ($[\mathbf{n}\cdot\boldsymbol{\xi}_\perp]_{S_p} \neq 0$) or *internal* ($[\mathbf{n}\cdot\boldsymbol{\xi}_\perp]_{S_p} = 0$) mode, corresponding to whether the plasma boundary does or does not shift during the perturbation, respectively. These two kinds of modes can equivalently be referred to as *free-boundary* or *fixed-boundary*, respectively. Secondly, we observe that the only terms that are not necessarily positive in δW_F are pressure-driven and current-driven, so the instability is classified according to its dominating driving factor. Current-driven modes are also frequently referred to as *kink modes*.

VI.II Stability in stellarators

VI.II.i Overview

Due to their three-dimensional structure, analysis of magnetohydrodynamic (MHD) stability in stellarator equilibria is an incredibly challenging problem. Current figures of merit used in stellarator optimization, such as the existence of a magnetic well, are frequently neither necessary nor sufficient for MHD stability [7]. While these objectives can provide physical insight into the stability properties of an equilibrium, more precise stability computations are necessary for improved stellarator optimization. Codes such as TERPSICHORE [15] and CAS3D [16] compute the linear instability growth rates of an equilibrium numerically, but these codes are time-intensive and must be called multiple times in order to obtain gradient information for one optimization step. Furthermore, both are legacy codes and require specific input and output formats, introducing discretization error, and neither are open-source, making 3D stability calculations significantly less accessible. A stability solver directly in the DESC optimization suite would enable autodifferentiation of stability calculations, dramatically reducing runtime and improving accuracy, while also being the first open-source MHD stability eigenvalue solver [17].

VI.II.ii Numerical implementations of stability to internal modes

CAS3D and TERPSICHORE CAS3D and TERPSICHORE numerically solve the variational principle problem (VI.6) as an eigenvalue equation [15] [16]. The change in potential energy due the perturbation $\boldsymbol{\xi}$ is given by

$$W_F = \frac{1}{2}\iiint d^3r[|\mathbf{C}|^2 - \mathcal{A}(\boldsymbol{\xi}\cdot\nabla s)^2 + \gamma p(\nabla\cdot\boldsymbol{\xi})^2] \tag{VI.10}$$

where $s = \rho^2$ is the flux surface label used by the VMEC code [18], and

$$\mathbf{C} = \nabla\times(\boldsymbol{\xi}\times\mathbf{B}) + \frac{\mathbf{J}\times\nabla s}{|\nabla s|^2}\boldsymbol{\xi}\cdot\nabla s$$

$$\mathcal{A} = 2|\nabla s|^{-4}(\mathbf{J}\times\nabla s)\cdot(\mathbf{B}\cdot\nabla)\nabla s$$

The perturbed kinetic energy (always positive) is given by

$$W_K = \frac{1}{2}\iiint d^3r\rho|\boldsymbol{\xi}|^2$$

With ρ the equilibrium mass density.

The additional constraint $\boldsymbol{\xi}\cdot\nabla s = 0$ is also enforced on the plasma boundary, since only fixed boundary modes are considered. By approximating $\boldsymbol{\xi}$ using a finite basis of linearly independent functions with coefficients $\mathbf{x}$, this becomes a matrix eigenvalue problem

$$P\mathbf{x} = \lambda K\mathbf{x}$$

Which can be solved numerically.

AGNI A current branch of DESC [19], currently under development by Rahul Gaur, has implemented numerical computations of stability to internal modes using similar principles. The code, hereafter referred to as AGNI (Analysis of Global Normal-modes in Ideal MHD), builds the full matrix A such that

$$\delta W_F = \mathbf{x}^T \mathrm{A} \mathbf{x}$$

Radial, poloidal, and toroidal derivatives of $\boldsymbol{\xi}$ are evaluated using pseudospectral differentiation matrices [20] [21], with the radial direction approximated using Chebyshev polynomials on the Chebyshev–Lobatto grid points, and the poloidal and toroidal directions approximated using Fourier series on a uniformly spaced grid.

For axisymmetric devices, in which the Fourier modes in the toroidal direction decouple, an option also exists to use only one grid point in the toroidal direction and set the toroidal differentiation matrix equal to $\mathrm{D}_\zeta = in$, where n is the toroidal mode number of the perturbation, provided as an input parameter by the user. This enforces $\boldsymbol{\xi}(\rho, \theta, \zeta) = \boldsymbol{\xi}(\rho, \theta) e^{in\zeta}$.

The Dirichlet boundary condition $\xi^\rho(\rho = 1, \theta, \zeta) = 0$ is enforced by removing the rows and columns of the matrix corresponding to ξ^ρ at the boundary.

For each grid point, a 3×3 symmetric positive semi-definite matrix B is also defined, which relates the vector components of $\mathbf{x}$ to the magnitude in real space of the displacement $\boldsymbol{\xi}$. B is then decomposed into a lower-triangular matrix and its transpose using Cholesky decomposition, i.e. $\mathrm{B}=\mathrm{L}\mathrm{L}^T$. The eigenvalues and eigenvectors of $\mathrm{L}^{-1}\mathrm{A}\mathrm{L}^{-T}$ are then calculated using standard methods for Hermitian matrices [22] [23] to solve the generalized eigenvalue problem $\lambda \mathrm{B}\mathbf{x} = \mathrm{A}\mathbf{x}$. The sign of the eigenvalue λ then corresponds to whether the equilibrium is stable to fixed-boundary modes.

VII Numerical implementations of MHD stability calculations in simple 1D models

In this section, the stability properties of two simple 1D models are explored numerically, as an illustration of the numerical methods used in the study of MHD stability.

VII.I Stability of the Z-pinch using the Energy Principle

VII.I.i Z-pinch

As a very simple example of a numerical implementation of the Energy Principle, the Energy Principle method was implemented for a simple one-dimensional case. The 1D Z-pinch is a one-dimensional model of a tokamak with a purely poloidal magnetic field. The model consists of an infinitely long cylinder extending in the z direction. Here, z is analogous to the toroidal direction, r is the flux surface label, and θ is the poloidal direction. All quantities are functions only of r. The plasma current $J_z(r)$ provides the magnetic field $B_\theta(r)$. The ideal MHD equilibrium equations for the Z-pinch reduce to:

$$\nabla \cdot \mathbf{B} = 0 \implies \frac{1}{r}\frac{\partial B_\theta}{\partial \theta} = 0$$

$$\mathbf{J} = \frac{1}{\mu_0}\nabla \times \mathbf{B} \implies J_z = \frac{1}{\mu_0 r}\frac{d}{dr}(rB_\theta)$$

$$\mathbf{J} \times \mathbf{B} = \nabla p \implies J_z B_\theta = -\frac{dp}{dr}$$

$$\mathbf{J} \times \mathbf{B} - \nabla p = 0 \implies -\frac{dp}{dr} = J_z B_\theta = \frac{B_\theta}{\mu_0 r}\frac{d}{dr}(rB_\theta) \implies \frac{d}{dr}\left(p + \frac{B_\theta^2}{2\mu_0}\right) + \frac{B_\theta^2}{\mu_0 r} = 0$$

Where the term in the derivative is the total particle and magnetic pressure, and the last term is the tension created by the magnetic curvature. The only free function is $B_\theta(r)$, from which $p(r)$ and $J_z(r)$ can be determined. The following example profiles satisfy the MHD equations for the Z-pinch:

$$\frac{2\mu_0 p(r)}{B_{\theta a}^2} = \frac{2}{3}(5 - 2\rho^2)(1 - \rho^2)^2$$

$$\frac{B_\theta(r)}{B_{\theta a}} = 2\rho(1 - \rho^2/2)$$

$$\frac{a\mu_0 J_z(r)}{B_{\theta a}} = 4(1-\rho^2)$$

Where $a = r(\rho = 1)$ is the radius of the cylinder and $B_{\theta a} = B_\theta(\rho = 1)$ is the field at the edge of the cylinder.

VII.I.ii Stability of the Z-pinch using the Energy Principle

The equilibrium is independent of θ, z, so the perturbation can be Fourier decomposed as $\boldsymbol{\xi}(r)e^{im\theta+ikz}$.

Treating the Z-pinch as equivalent to a torus with major radius R_0 and minor radius a, and defining $\xi \equiv \xi_r$ for notational convenience (since ξ_θ and ξ_z will be minimized with respect to ξ_r), the expression for δW_F can be derived:

$$\frac{\delta W_F}{2\pi R_0} = \pi \int_0^a W(r)dr$$

With

$$W(r) = \frac{m^2 B_\theta^2}{\mu_0 r^2}(|\xi|^2+|\xi_z|^2)+\frac{B_\theta^2}{\mu_0}\left|r(\xi/r)' + ik\xi_z\right|^2+\frac{2p'}{r}|\xi|^2+\gamma p\left|\frac{1}{r}(r\xi)' + \frac{im\xi_\theta}{r} + ik\xi_z\right|^2 \tag{VII.1}$$

For $m \neq 0$, ξ_θ, ξ_z, and k can be chosen to minimize $W(r)$, such that

$$W(r) = \left(\frac{2p'}{r} + \frac{m^2 B_\theta^2}{\mu_0 r^2}\right)|\xi|^2 r$$

So that a necessary and sufficient condition for $\delta W_F > 0$, $m \neq 0$ is

$$rp' + \frac{m^2 B_\theta^2}{2\mu_0} > 0 \Leftrightarrow \frac{1}{B_\theta^2}(rB_\theta^2)' < m^2 - 1$$

For a standard Z-pinch, $B_\theta \propto r$ near the origin by Ampere's law, so that the $m = 1$ mode is never stable. For $m = 0$, ξ_θ cannot be chosen to minimize the compressibility term, so the stability condition becomes

$$-\frac{rp'}{p} < \frac{2\gamma B_\theta^2}{B_\theta^2 + \mu_0\gamma p}$$

At large radius in a well-confined plasma, $p \ll \frac{B_\theta^2}{\mu_0}$ so the criterion becomes $-\frac{rp'}{p} < 2\gamma$, imposing very stringent conditions on the pressure profile. Therefore the Z-pinch is always unstable to the $m = 1$ mode and frequently unstable to the $m = 0$ mode.

VII.I.iii Numerical implementation

Energy Principle Method The perturbation $\boldsymbol{\xi}$ can be described by N Chebyshev basis modes evaluated at M real nodes. Define $\mathbf{x}$ to be the $3N$-component vector containing the flattened Chebyshev modes for ξ_r, ξ_θ, ξ_z. Now let I_r, I_θ, I_z be the $3N \times N$ matrices that identify the r, θ, and z spectral coefficients of $\boldsymbol{\xi}$, respectively. Let A be the $N \times M$ matrix of the N spectral modes at the M output nodes, and define D to be the $N \times M$ matrix containing the radial derivatives of the basis functions. In particular, this means the vector ξ_i that gives the values of $\boldsymbol{\xi}(i = r, \theta, z)$ at the M real nodes is given by $\xi_i = AI_i\mathbf{x}$, $\xi_i' = DI_i\mathbf{x}$. Finally, let C be the diagonal $N \times N$ matrix encoding the weights of each real node.

Replacing all $|...|^2$ by $(...)^\dagger(...)$, and all ξ_r and $\xi_r^\dagger$ in (VII.1) using the above formulas, an expression in the form of $\mathbf{x}^T A\mathbf{x}$ can be obtained, where A is a real symmetric matrix. This matrix can then be eigendecomposed in order to find the least stable eigenvalue.

Comparison with analytic formulas Setting $\gamma = 2$ for simplicity, the generic profiles set by

$$
\begin{aligned}
p(\rho) &= \frac{p_0}{(1+\rho^2)^\nu} \\
p'(\rho) &= -\frac{2p_0\nu\rho}{(1+\rho^2)^{\nu+1}} \\
B_\theta^2(\rho) &= \frac{2p_0}{\nu-1}\frac{1}{\rho^2}\left[1 - \frac{(1+\nu\rho^2)}{(1+\rho^2)^\nu}\right]
\end{aligned}
$$

can be used. The stability boundary corresponds to $\nu = 2$.

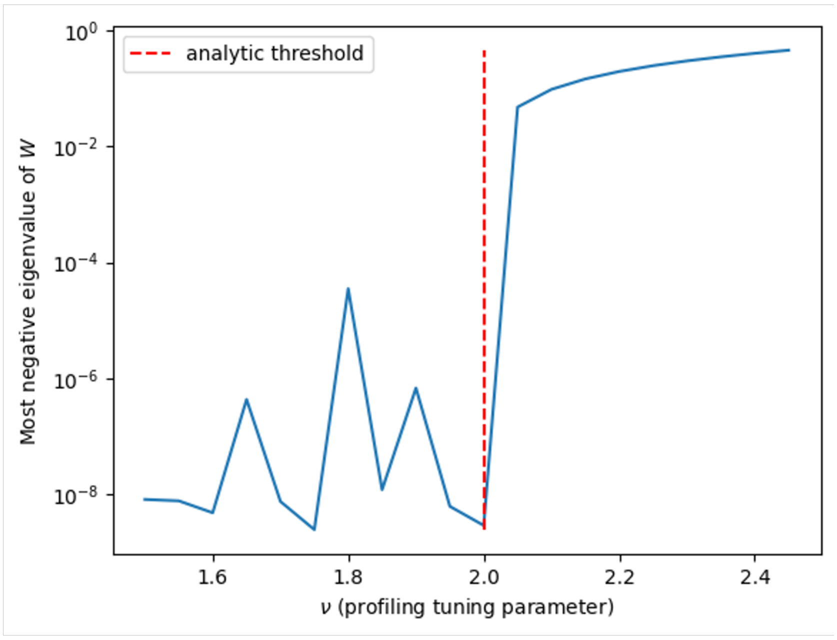

Figure 3: Magnitude of the most negative eigenvalue of the Z-pinch, with the profiles given by (VII.2). Although the smallest eigenvalue is never positive, there is a jump by multiple orders of magnitude when crossing the analytic stability boundary.

VII.II Newcomb solver for a straight tokamak in DESC

Very few analytic results exist for the stability of stellarators, due to their highly complex 3D geometry. Instead, for validation of AGNI, the stability solver described in VI.II.ii, and its extension to external modes, as described in VIII, the stability solver is used to evaluate the stability boundary of a very high-aspect ratio circular tokamak, the simplest geometry that can be described by DESC. In the limit of small inverse aspect ratio, defined as $\varepsilon = \frac{a}{R_0} \to 0$, this device can be treated as a screw pinch, which is a one-dimensional model of a tokamak with nonzero poloidal and toroidal magnetic fields. In this case, analytic results are possible and very useful for validation — the results and comparison to results from the stability solver are detailed in IX. In order to illustrate the stability properties of a screw pinch, a script to evaluate Newcomb's procedure on a straight tokamak DESC equilibrium has been implemented.

At equilibrium, the pressure profile and magnetic fields in a screw pinch satisfy

$$\frac{d}{dr}\left(p + \frac{B_z^2}{2\mu_0}\right) + \frac{B_\theta}{\mu_0 r}\frac{d}{dr}\left(rB_\theta\right) = 0 \tag{VII.3}$$

With $\hat{r}, \hat{\theta}, \hat{z}$ a right-handed orthonormal coordinate system ($\hat{r} \times \hat{\theta} = \hat{z}$).

VII.II.i Overview of Newcomb's procedure

Newcomb's procedure is a procedure that can be used to compute the marginal stability boundary of a screw pinch as follows. First, exploiting the cylindrical symmetry, the perturbation $\boldsymbol{\xi}$ can be Fourier decomposed as $\boldsymbol{\xi}(\mathbf{r}) = \boldsymbol{\xi}(r)\exp(im\theta + ikz)$. Then defining

$$F(r) \equiv \mathbf{k} \cdot \mathbf{B} = kB_z + mB_\theta/r \tag{VII.4a}$$

$$G(r) \equiv \mathbf{e}_r \cdot \mathbf{k} \times \mathbf{B} = mB_z/r - kB_\theta \tag{VII.4b}$$

Newcomb's procedure for internal modes can be implemented as follows:

1. First, evaluate Suydam's criterion, a necessary but not sufficient criterion for stability against pressure-driven localized interchange instabilities. The criterion describes a competition between the stabilizing effects of line bending and the destabilizing effect of unfavorable curvature. The criterion is given by

$$rB_z^2\left(\frac{q'}{q}\right)^2 + 8\mu_0 p' > 0$$

   Where $q(r) = \frac{rB_z}{R_0 B_\theta} = \frac{1}{\iota(r)}$.

2. If Suydam's criterion is not violated, test for stability to internal modes.

   (a) If $F(r) \neq 0$ for all $r \in (0, a)$, then the following procedure determines stability:

      i. Defining $\xi = \boldsymbol{\xi} \cdot \mathbf{e}_r$ and minimizing δW_F with respect to the other two components of the perturbation, the perturbed plasma potential energy from a fixed-boundary mode is given by

$$\frac{\delta W_F}{2\pi^2 R_0/\mu_0} = \int_0^a \left(f\xi'^2 + g\xi^2\right) dr$$

      with

$$k_0^2 = k^2 + m^2/r^2$$

$$f(r) = \frac{rF^2}{k_0^2}$$

$$g(r) = 2\mu_0\frac{k^2}{k_0^2}p' + \frac{k_0^2 r^2 - 1}{k_0^2 r^2} rF^2 + 2\frac{k^2}{rk_0^4}FF^\dagger$$

for $F^\dagger = kB_z - \frac{mB_\theta}{r}$. So, the perturbation that minimizes δW_F satisfies

$$\frac{d}{dr}\left(f\frac{d\xi}{dr}\right) - g\xi = 0 \tag{VII.5}$$

ii. There are two linearly independent solutions to (VII.5). Without loss of generality, there exists a nontrivial solution ξ_1 that is regular at $r = 0$. Near $r = 0$, $f(r) \sim \frac{F^2}{m^2}r^3$ and $g(r) \sim \frac{F^2}{m^2}(m^2-1)r$. Letting $\xi \sim r^p$, we find $p = -1 \pm m$, so $\xi_1(r) \sim r^{|m|-1}$ near the axis. Alternatively, if $m = 0$, then $f \sim \frac{rF^2}{k^2}$ and $g \sim \frac{F^2}{rk^2}$ near axis, and so $\xi_1 \sim r$ near axis. (VII.5) can then be integrated to yield ξ_1 at all r. If $\xi_1(r)$ has a zero crossing in $(0, a)$, the pinch is unstable to internal modes.

(b) Else, divide the screw pinch into radial regions where $F(r) \neq 0$. Repeat the previous analysis for each sub-interval. However, for subintervals (r_s, r_{s+1}) that do not begin at $r = 0$, the power law for ξ_1 near r_s is given by

$$\xi_1 \sim (r - r_s)^{p_1}\ ,\ p_1 = -\frac{1}{2} + \frac{1}{2}(1 - 4D_S)^{\frac{1}{2}}\ ,\ D_S = -\left(\frac{2\mu_0 p' q^2}{rB_z^2 q'^2}\right)_{r_s}$$

Suydam's criterion is equivalent to $1-4D_S > 0$, so p_1 is real. As before, if ξ_1 has a zero crossing in (r_s, r_{s+1}), then the plasma is unstable.

A proof of this procedure's validity is given in Freidberg [7]. Observe also that the only m, k wavenumbers that must be examined are $m = 1$, $-\infty < k < \infty$ and $m = 0$, $k^2 \to 0$, since these modes will always be the least stable. Since the device considered here is a high aspect ratio tokamak, enforcing toroidal periodicity on the perturbation means that $k = -\frac{n}{R_0}$ for toroidal mode number $n \in \mathbb{Z}$. Figures 4 and 5 demonstrate the behavior of different eigenmodes near the associated rational surfaces.

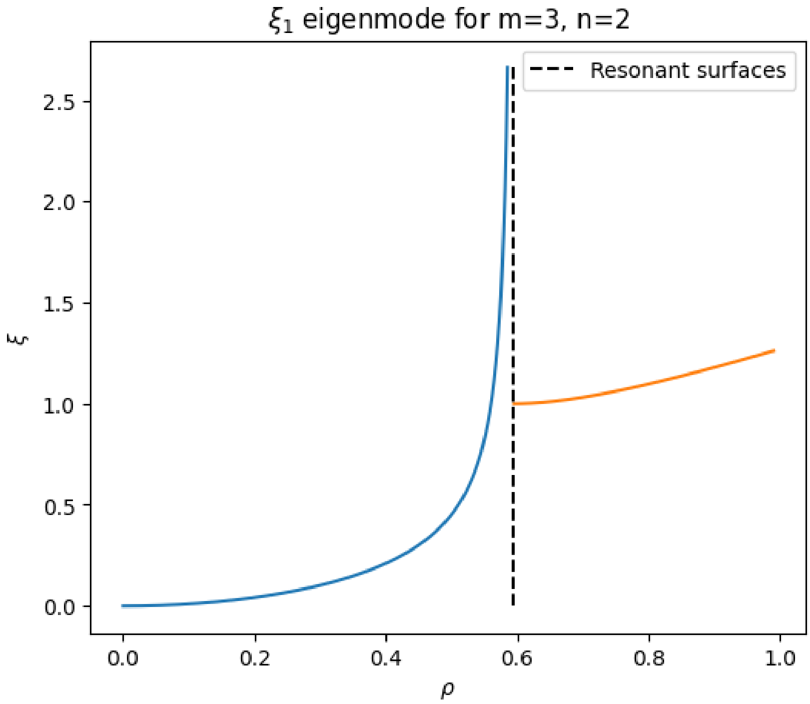

Figure 4: Solution to the minimizing differential equation in a screw pinch equilibrium for the $m = 3$, $n = 2$ mode.

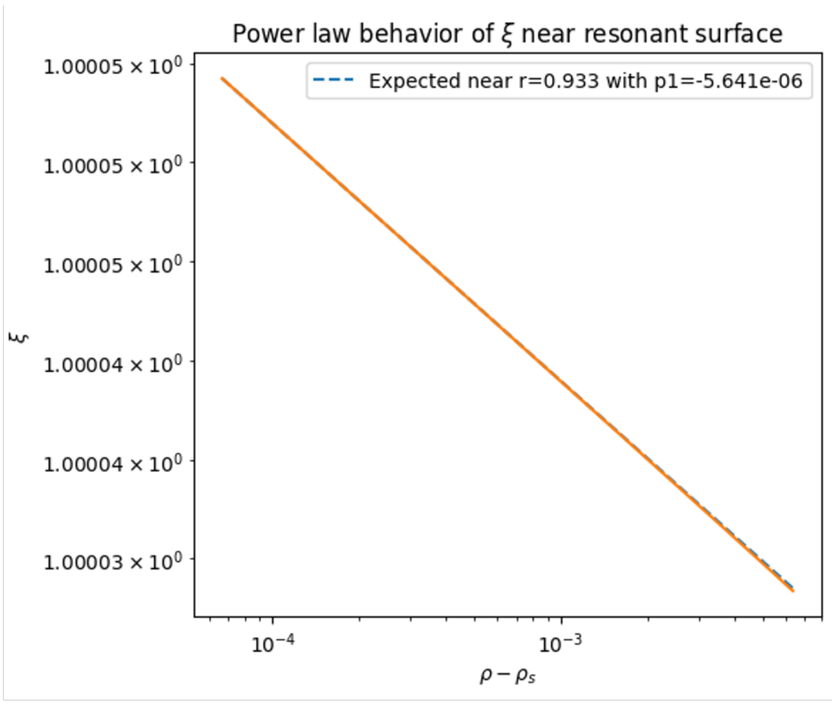

Figure 5: Close alignment between the $m = 1$, $n = 1$ and the expected power law behavior near the $\iota = 1$ resonant surface in a screw pinch equilibrium.

VIII Implementation of external modes in the DESC stability solver

As described in VI.II.ii, AGNI, a solver that computes stability to internal/fixed boundary modes, already is in development [24]. The goal of this work is to extend AGNI to external modes.

Merkel et al. [25] extends the CAS3D code to consider external modes as well; this section generally follows the method used in that work.

The perturbed vacuum energy term is given by

$$\delta W_V = \frac{1}{2\mu_0} \iiint_V d^3\mathbf{r} |\mathbf{B}_{V_1}|^2$$

where $\mathbf{B}_{V_1}$ is the first-order perturbed magnetic field in the vacuum region.

DESC assumes that the total pressure is continuous across the plasma boundary, i.e. $\left[\left[p + \frac{1}{2}B_p^2 - \frac{1}{2}B_v^2\right]\right] = 0$, so that the surface energy vanishes; $\delta W_S = 0$, and the only term that needs to be considered in the extension of AGNI to external modes is δW_V.

The perturbed vacuum magnetic energy satisfies

$$\nabla \cdot \mathbf{B}_{V_1} = 0 \qquad\qquad \nabla \times \mathbf{B}_{V_1} = 0$$

So that there exists a scalar potential Φ with $\mathbf{B}_{V_1} = \nabla\Phi$.

Since Φ satisfies Laplace's equation in the vacuum, we can evaluate

$$\begin{aligned}
\delta W_V &= \frac{1}{2\mu_0} \iiint_V d^3\mathbf{r} |\mathbf{B}_{V_1}|^2 && \text{(VIII.1a)}\\
&= \frac{1}{2\mu_0} \iiint_V d^3\mathbf{r} |\nabla\Phi|^2 && \text{(VIII.1b)}\\
&= \frac{1}{2\mu_0} \iint_S d\mathbf{S} \cdot (\Phi\nabla\Phi) && \text{(VIII.1c)}\\
&= \frac{1}{2\mu_0} \iint_S dS \left[\Phi\partial_n\Phi\right] && \text{(VIII.1d)}
\end{aligned}$$

Where the third equality follows from integration by parts and $\nabla^2\Phi = 0$.

In this section, the magnetic field in the plasma will be denoted by $\mathbf{B}_P$ to distinguish the magnetic field in the plasma volume from the field in the vacuum region. By the divergence-free condition $\nabla \cdot \mathbf{B} = 0$, the perturbed magnetic field must be continuous at the plasma boundary: $\mathbf{B}_{V_1} \cdot \hat{\mathbf{n}} = \mathbf{B}_{P_1} \cdot \hat{\mathbf{n}}$. Therefore, the scalar potential Φ must satisfy the boundary condition

$$\begin{aligned}
\partial_n \Phi &\equiv \mathbf{n} \cdot \nabla\Phi && \text{(VIII.2a)}\\
&= \mathbf{n} \cdot \mathbf{B}_{P_1} && \text{(VIII.2b)}\\
&= \mathbf{n} \cdot \nabla \times (\boldsymbol{\xi} \times \mathbf{B}_P) && \text{(VIII.2c)}\\
&= |\nabla\rho|^{-1}\mathbf{B}_P \cdot \nabla\xi^\rho && \text{(VIII.2d)}
\end{aligned}$$

Where the third line follows from (VI.2), and the fourth line follows from the use of magnetic coordinates, in which $\mathbf{B}_p^\rho = 0$ and $\sqrt{g}_m \mathbf{B}_P^\vartheta$ and $\sqrt{g}_m \mathbf{B}_p^\zeta$ are functions only of ρ.

Finally, in magnetic coordinates, we can use (V.4e) to write

$$\begin{aligned}
\mathbf{B}_p \cdot \nabla\xi^\rho &= \frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m}(\mathbf{e}_\zeta + \iota\mathbf{e}_\vartheta) \cdot \nabla\xi^\rho && \text{(VIII.3a)}\\
&= \frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m}\left(\frac{\partial\xi^\rho}{\partial\zeta} + \iota\frac{\partial\xi^\rho}{\partial\vartheta}\right) && \text{(VIII.3b)}
\end{aligned}$$

We can therefore rewrite (VIII.1) as

$$\begin{aligned}
\delta W_V &= \frac{1}{2\mu_0}\iint_S d\mathbf{S} \cdot (\Phi\nabla\Phi) && \text{(VIII.4a)}\\
&= \frac{1}{2\mu_0}\iint_S d\theta d\zeta \sqrt{g}_m \Phi \left[e^\rho \cdot \nabla\Phi\right] && \text{(VIII.4b)}\\
&= \frac{1}{2\mu_0}\iint_S d\theta d\zeta \sqrt{g}_m \Phi \left[\mathbf{B}_p \cdot \nabla\xi^\rho\right] && \text{(VIII.4c)}\\
&= \frac{1}{2\mu_0}\iint_S d\theta d\zeta \Phi \left[\psi'(\rho)|_{\rho=1}\left(\frac{\partial\xi^\rho}{\partial\zeta} + \iota\frac{\partial\xi^\rho}{\partial\vartheta}\right)\right] && \text{(VIII.4d)}
\end{aligned}$$

Defining $G(\mathbf{r}, \mathbf{r}') = \frac{1}{|\mathbf{r}-\mathbf{r}'|}$, which is the Green's function of Laplace's equation ($\nabla^2 G(\mathbf{r}, \mathbf{r}') = -4\pi\delta(\mathbf{r} - \mathbf{r}')$), we can use Green's second identity to write

$$\begin{aligned}
&\iiint_V \left[\Phi(\mathbf{r}')\nabla^2 G(\mathbf{r}, \mathbf{r}') - G(\mathbf{r}, \mathbf{r}')\nabla^2\Phi(\mathbf{r}')\right] d^3\mathbf{r}'\\
&= \iint_S d\mathbf{S} \cdot (\Phi(\mathbf{r}')\nabla' G(\mathbf{r}, \mathbf{r}') - G(\mathbf{r}, \mathbf{r}')\nabla'\Phi(\mathbf{r}'))
\end{aligned}$$

So that

$$\iint_S d\mathbf{S}' \cdot (G(\mathbf{r},\mathbf{r}')\nabla'\Phi(\mathbf{r}')) = 2\pi\Phi(\mathbf{r}) + \iint_S d\mathbf{S}' \cdot (\Phi(\mathbf{r}')\nabla' G(\mathbf{r},\mathbf{r}')) \qquad \text{(VIII.5)}$$

Where $d\mathbf{S}\cdot\nabla\Phi(\mathbf{r}) = |\nabla\rho|^{-1}\mathbf{B}_p\cdot\nabla\xi^\rho dS$, as in (VIII.2), so that the right side of the equation can be computed from the perturbation and from equilibrium quantities.

When discretized, (VIII.5) becomes a linear equation in Φ, which can be solved to obtain

$$\vec{\Phi} = M_\Phi \vec{B}_{n_1}$$

for some matrix M_Φ. The method used to find M_Φ is described in VIII.I.

$\vec{B}_{n_1}$ is itself linearly related to $\boldsymbol{\xi}$ by (VIII.2) and (VIII.3):

$$\begin{aligned}
\vec{B}_{n_1} &= \frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m|\nabla\rho|}\left(\frac{\partial\vec{\xi}^\rho}{\partial\zeta} + \iota\frac{\partial\vec{\xi}^\rho}{\partial\vartheta}\right) && \text{(VIII.6a)}\\
&= \frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m|\nabla\rho|}\left(D_\zeta + \iota D_\vartheta\right)\vec{\xi}^\rho && \text{(VIII.6b)}\\
&\equiv \frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m|\nabla\rho|}D_{\mathbf{B}}\vec{\xi}^\rho && \text{(VIII.6c)}
\end{aligned}$$

Here $D_\vartheta \equiv \frac{\partial}{\partial\vartheta}$ and $D_\zeta \equiv \frac{\partial}{\partial\zeta}$ are discrete differentiation matrices, and $\frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m|\nabla\rho|}$ is diagonal.

This then implies

$$\vec{\Phi} = M_\Phi \vec{B_{n,1}} = M_\Phi\frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m|\nabla\rho|}D_{\mathbf{B}}\vec{\xi}^\rho$$

Overall, then, we have

$$\begin{aligned}
\delta W_V &= \frac{1}{2\mu_0}\iint_S d\theta d\zeta\Phi\left[\psi'(\rho)|_{\rho=1}\left(\frac{\partial\xi^\rho}{\partial\zeta} + \iota\frac{\partial\xi^\rho}{\partial\vartheta}\right)\right] && \text{(VIII.7a)}\\
&= \Phi^\dagger W\psi'(\rho)|_{\rho=1}D_{\mathbf{B}}\vec{\xi}^\rho && \text{(VIII.7b)}\\
&= \frac{1}{2\mu_0}\vec{\xi}^{\rho\dagger}\left(M_\Phi\frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m|\nabla\rho|}D_{\mathbf{B}}\right)^\dagger C\psi'(\rho)|_{\rho=1}D_{\mathbf{B}}\vec{\xi}^\rho && \text{(VIII.7c)}
\end{aligned}$$

Where C is a diagonal matrix consisting of integration weights, and $\vec{\xi}^\rho$ is evaluated on the boundary.

So the matrix representing the vacuum energy is

$$\delta W_V = \boxed{\frac{1}{2\mu_0}\left(M_\Phi \frac{\psi'(\rho)|_{\rho=1}}{\sqrt{g}_m |\nabla\rho|} D_{\mathbf{B}}\right)^\dagger C\psi'(\rho)|_{\rho=1} D_{\mathbf{B}}} \tag{VIII.8}$$

Explicitly, the expression used in the code is given by

```
A = A.at[b_idx, b_idx].add( # indices corresponding to xi^rho on the boundary
    # _cT is the conjugate transpose
    -_cT(
        # C has shape (3*n_total, 1); equiv to @ by diag matrix on the left
        C[b_idx, :] # Integration weights
        * psi_r_s**3 # this is just for consistency; psi' = 1 here
        * (iota_s * D_theta[b_idx, b_idx] + D_zeta[b_idx, b_idx]) # || derivative
    )
    @ (
        phi_matrix # M_Phi
        @ (
            # Bdotn
            psi_r_s
            / sqrtg_grad_rho
            * (iota_s * D_theta[b_idx, b_idx] + D_zeta[b_idx, b_idx])
        )
    )
)
```

Here, all quantities are normalized by a_N, the average minor radius, and $B_N = \frac{\Psi(a)}{\pi a_N^2}$ to be dimensionless, leading to an overall normalization of $\frac{1}{B_N a_N^3/2\mu_0}$. Additionally, AGNI defines $\xi^\rho = \frac{d\tilde{\psi}}{d\rho}\tilde{\xi}^\rho$ with $\frac{d\tilde{\psi}}{d\rho} = \rho$, and then solves for $\tilde{\xi}^\rho$, in order to enforce $\xi^\rho = 0$ on axis. On the boundary, $\frac{d\tilde{\psi}}{d\rho} = 1$, so this additional line is simply for consistency with all other terms in the matrix. In the case in which the Dirichlet boundary condition is enforced ($\xi^\rho(1,\theta,\zeta) = 0$), these lines of the matrix are removed entirely.

VIII.I Exterior Neumann problem

As discussed in section VIII, solving Laplace's equation in the vacuum region outside the plasma, subject to a fixed $\mathbf{B}\cdot\hat{\mathbf{n}}$ on the boundary, is necessary to compute the perturbed vacuum energy δW_V. This problem can be reduced to the boundary integral equation (VIII.5). The goal is to obtain the matrix M_Φ encoding the linear dependence of Φ on $\mathbf{B} \cdot \hat{\mathbf{n}}$; in order to achieve this goal, both singular integrals in (VIII.5) must be computed on a representative set of basis vectors. Since the discrete Fourier transform is also a linear operator, this is equivalent to evaluating Φ from $\mathbf{B} \cdot \hat{\mathbf{n}}$ directly.

In other words, the discretization in spectral space of the following two linear operators must be computed:

$$D[f](\mathbf{r}) = \frac{1}{4\pi} \iint d\mathbf{S}' \cdot (f(\mathbf{r}')\nabla' G(\mathbf{r}, \mathbf{r}')) - \frac{f}{2} \tag{VIII.9a}$$

$$S[f](\mathbf{r}) = \frac{1}{4\pi} \iint dS' \left(f(\mathbf{r}') G(\mathbf{r}, \mathbf{r}')\right) \tag{VIII.9b}$$

So that the exterior Neumann problem (VIII.5) can be written more compactly as

$$S[\mathbf{B} \cdot \hat{\mathbf{n}}] = D[\Phi] \tag{VIII.10}$$

Merkel [26] handles the singularity in the Green's function $G(\mathbf{r}, \mathbf{r}')$ in the integral using a singularity subtraction scheme: the work defines analytic expressions with the same singular behavior to those of the kernels, such that the nonsingular component of the kernels can be integrated numerically and the singular components can be integrated analytically. This technique is used in CAS3D for external mode stability calculations [25] and in the NESTOR code for free-boundary equilibrium calculations [26]. However, this method is accurate only to second order in the number of grid points used to discretize the singular integral [27].

In order to compute free-boundary equilibria, DESC uses the technique developed by Malhotra et al. [28] [29] to compute the singular virtual casing integral; this method is accurate to 10th to 12th order in the number of grid points. The method is as follows:

1. First, define the partition of unity η. The partition of unity is only nonzero in an M×M array of source grid points around the singularity, which is located at (θ_0, ζ_0) in flux coordinates.

$$\eta(\theta, \phi) = \chi\left(\frac{2}{M}\sqrt{\left(\frac{\theta - \theta_0}{h_\theta}\right)^2 + \left(\frac{\zeta - \zeta_0}{h_\zeta}\right)^2}\right)$$

Here, $\chi(r) = e^{-36|r|^8}$, and h_θ and h_ζ are the spacing of the source grid points in θ and ζ, respectively.

2. The next step is to evaluate the nonsingular integral, which is the original integrand, but with the integrand multiplied by $(1-\eta)$.

3. Then, a new polar coordinate system is defined with respect to the original coordinate system, i.e. r and ω defined such that $\theta = \theta_0 + \frac{M}{2} h_\theta r \sin\omega$ and $\zeta = \zeta_0 + \frac{M}{2} h_\zeta r \cos\omega$.

4. A new quadrature grid must then be defined in terms of the new polar coordinate system, and the relevant input data for the integrand (e.g. K_{vc}, the virtual casing sheet current) must be interpolated to this new grid. The singular part of the integral (the original integral, multiplied by $\eta(r)$ and integrated in the region where $r \leq 1$) can then be integrated on the new quadrature grid.

5. The singular and nonsingular parts of the integral are then added together to compute the total integral.

Unalmis [30] uses this method of computing singular integrals to implement the solution to the exterior Neumann problem in DESC. That work constructs the interpolator necessary to compute the singular integral, and then discretizes the kernel (VIII.9a) as a matrix acting on Φ, while computing (VIII.9b) for arbitrary f. This work extends that work to compute the full discretization of $S[f]$ as well. Specifically, both $D[f]$ and $S[f]$ are evaluated on a two-dimensional basis set of Fourier modes.

The linear system of equations (VIII.10) is then solved in Fourier space. The spectral matrix is then composed with the discrete Fourier transform and discrete inverse Fourier transform to obtain M_Φ in real space, i.e.

$$M_\Phi = FD^{-1}SF^{-1}$$

where F represents the discrete Fourier transform.

This work also adds an option to evaluate both kernels (VIII.9) on a uniform grid in PEST coordinates, rather than an assumed uniform grid in (ρ, θ, ζ) coordinates. For the PEST implementation, the DESC coordinates must also be provided in order to compute surface quantities. The PEST coordinate implementation of M_Φ is necessary for the extension of AGNI to external modes.

VIII.II Validation of solution to exterior Neumann problem

VIII.II.i Recreation of toy vacuum field in a non-axisymmetric geometry

Of critical importance in the computation of the vacuum energy δW_V is accurate evaluation of the scalar magnetic potential Φ on the boundary. Therefore, it is useful to determine the resolution necessary to accurately reproduce the magnetic field on the boundary. The example magnetic field was defined to be

$$\mathbf{B} = \frac{1}{4\pi}\nabla\left[\frac{1}{|\mathbf{x}-\mathbf{x}_0|}\right] = \frac{1}{4\pi}\frac{\mathbf{x}-\mathbf{x}_0}{|\mathbf{x}-\mathbf{x}_0|^3}$$

where $\mathbf{x}_0$ is the location of the magnetic axis at $\zeta = 0$. This magnetic field was used to test the original implementation of the exterior Neumann problem in Unalmis [30].

While this choice of magnetic field is not physical — it is not divergence free at $\mathbf{x} = \mathbf{x}_0$ — it satisfies $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{B} = 0 \; \forall \; \mathbf{x} \in \mathbf{R}^3 \setminus \{\mathbf{x}_0\}$. This choice of field is therefore convenient, since for any surface $\mathcal{S} \subset \mathbf{R}^3 \setminus \{\mathbf{x}_0\}$, the solution Φ to the exterior Neumann problem must satisfy $\Phi(\mathbf{x}) = \frac{1}{4\pi}\frac{1}{|\mathbf{x}-\mathbf{x}_0|}$ up to an additive constant. However, this choice of magnetic field may tend to overestimate the number of Fourier modes necessary to accurately compute Φ, as the singularity at $\mathbf{x}_0$ could lead to slower spectral convergence.

$\mathcal{S}$ was chosen to be the plasma boundary of the planned stellarator NCSX [31], in order to verify the implementation of (VIII.10) in a non-axisymmetric geometry. The PEST coordinates from that equilibrium were also used to solve Laplace's equation, in order to validate the implementation of (VIII.10) in PEST coordinates.

The maximum poloidal mode number M and maximum toroidal mode number N was varied, leading to a total of (2M+1)(2N+1) Fourier modes. The collocation grid consisted of 2M and 2N grid points uniformly spaced in the ϑ and ζ directions, respectively. Three different resolution scans were performed:

1. N=M
2. N=2M
3. N=$\lfloor \frac{R_0}{a} \mathrm{M} \rfloor$

As seen in Figure 6, at sufficiently high resolution, all three scans converge to $\Phi(\mathbf{x})$, as desired.

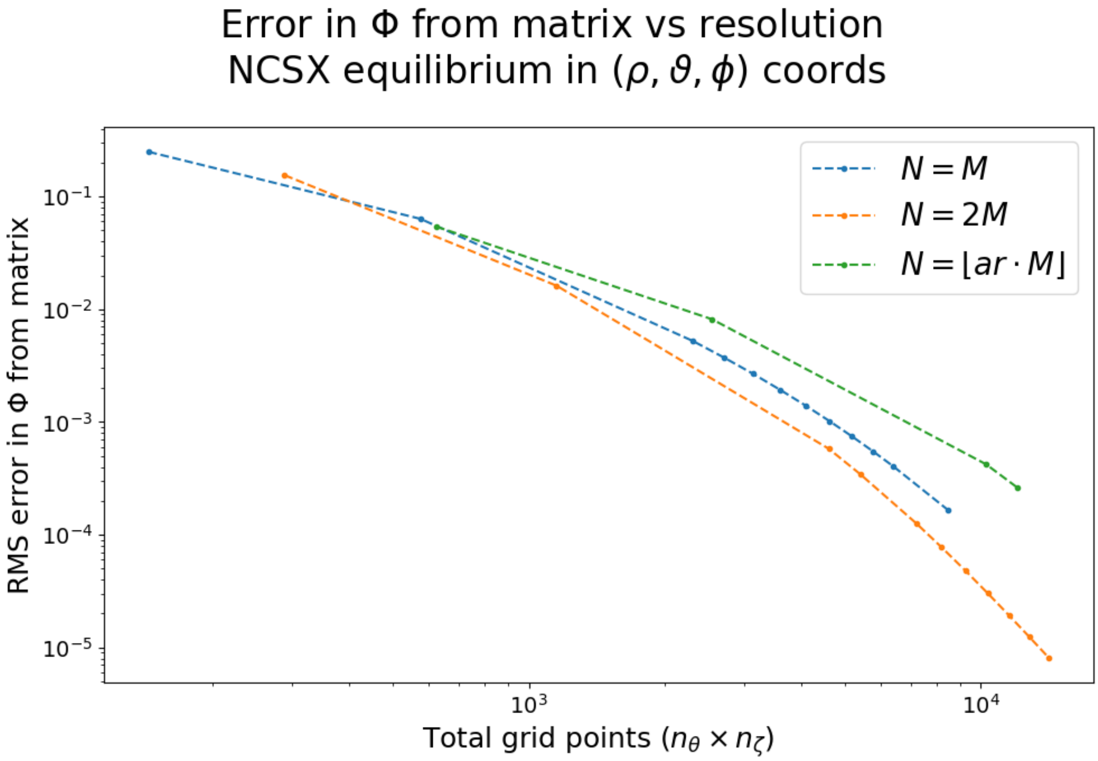

Figure 6: Convergence of $M_\Phi \vec{B}_n$ to $\frac{1}{4\pi}\frac{1}{|\mathbf{x}-\mathbf{x}_0|}$ as the number of grid points increases. All three resolution scans accurately solve the exterior Neumann problem in a non-axisymmetric geometry.

VIII.II.ii Accurate integration of vacuum energy

The second numerical test of M_Φ was to validate that $\vec{B}_n\sqrt{g}_m|\nabla\rho|CM_\Phi\vec{B}_n$ accurately estimates the vacuum energy W_V. Here, the choice of surface was a circular tokamak with major radius $R_0 = 4$ and minor radius $a = 1$. Two DESC coil objects were placed inside the tokamak at $(R_0, \phi, \pm\Delta_h)$, with opposing currents of $\pm 10^6$ A. This choice of magnetic field ensures that $\oint_{\mathcal{C}} \mathbf{B}\cdot d\boldsymbol{\ell} = \mu_0 I_{\mathrm{enc}} = 0$ for any poloidal or toroidal loop $\mathcal{C}$ outside the torus. Therefore, the scalar potential Φ has no secular terms and its Fourier decomposition is mathematically valid. Figure 7 depicts the test setup.

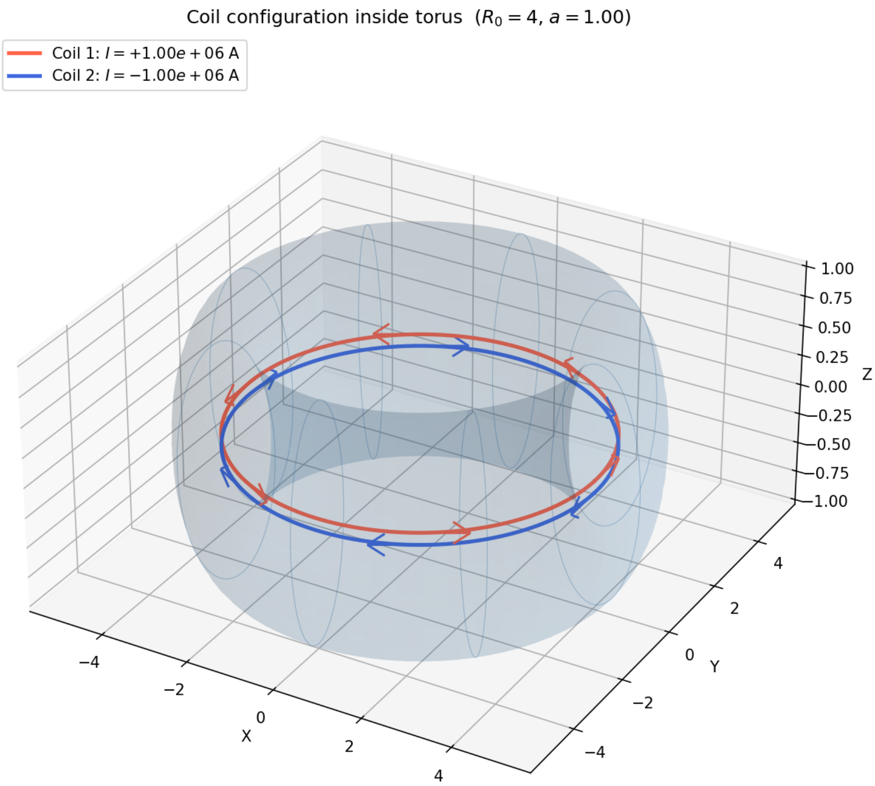

Figure 7: The coil configuration and surface used to test the estimate of the vacuum energy W_V. The surface is a circular torus, and the two coils, at $Z = \pm\Delta_h$, have equal and opposite currents.

The surface integral (VIII.1) obtained from M_Φ was then compared to an integral over a volume approximating the whole space, which was taken out to $R = 10000$ meters. Δ_h/a was varied from 0.01 to 0.1 in order to vary the angular dependence of $\mathbf{B} \cdot \hat{\mathbf{n}}$ on the surface, and the two methods for computing the vacuum energy were compared. Figure 8 demonstrates the close alignment between the direct volume integral and the equivalent surface integral, corresponding to a percentage error of $\sim$0.5%.

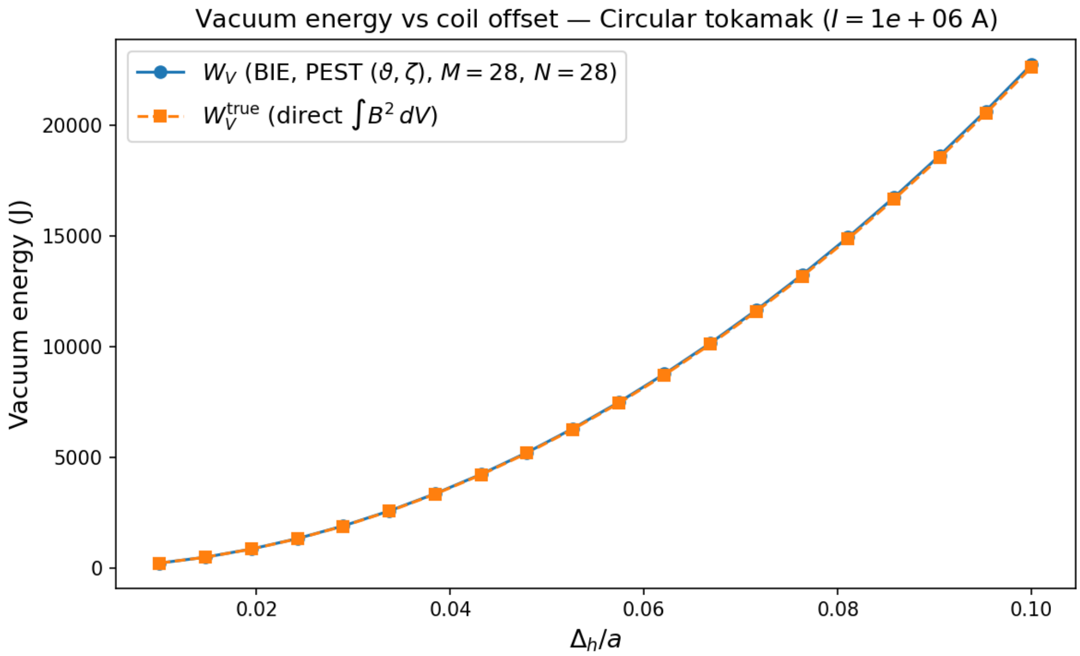

Figure 8: The vacuum magnetic energy produced by the two coils at varying coil separations Δ_h; the blue line corresponds to the vacuum energy as estimated by the surface integral (VIII.1) and the orange line corresponds to a direct volume integral of the field from the coils.

Together, the two test cases described in this section validate the implementation of M_Φ in PEST coordinates, including in non-axisymmetric geometries, and its use in integration of the vacuum energy.

IX Comparison to analytic results for a straight tokamak

As mentioned in VII.II, the straight tokamak is convenient for validation, as its simple geometry enables analytic progress. All analytic results in this section, unless otherwise stated, are proven in Chapter 11 of Freidberg [7].

IX.I Ordering

In the large aspect ratio limit for a circular tokamak, also known as the straight tokamak limit, some analytic results can be derived. In this limit, the following ordering is assumed, with $\varepsilon = \frac{a}{R_0} \ll 1$:

Large aspect ratio $\frac{a}{R_0} = \varepsilon$

Low pressure $\beta = \frac{p}{\frac{B^2}{2\mu_0}} \sim \varepsilon^2$

Rotational transform of order 1 $\frac{1}{q} = \iota \sim 1$

Note that as a consequence of $\frac{a}{R_0} \sim \varepsilon$ and $\iota \sim 1$, it follows that $\frac{B_\theta}{B_z} \sim \varepsilon$.

IX.II Analytic results and comparison to results from eigenvalue solver

For all test cases, a circular tokamak with an aspect ratio of $\frac{1}{\varepsilon} = \frac{R_0}{a} = 10$ was used. In order to maximize resolution in the radial and poloidal directions subject to memory constraints, only one toroidal grid point was used, and the toroidal mode number of the perturbation was fixed at $n = 1$ by defining $D_\zeta = i$, as described in VI.II.ii.

IX.III Internal $m = 1$ kink mode

IX.III.i Analytic results

In the case of internal modes, where $\xi^\rho(r = a, \theta, z) = 0$, the straight tokamak is stable to all modes except the $m = 1$ kink instability. The stability condition here requires that $\iota_0 < 1$, where ι_0 is the ι on the axis, and that $\iota(r)$ decreases with ρ.

In this case, the perturbed energy (to second order in ε) becomes

$$\delta\hat{W}_2 = \frac{\delta W_2}{\varepsilon^2 W_0} = \int_0^a \left(\frac{n}{m} - \frac{1}{q}\right)^2 \left[r^2\xi'^2 + (m^2-1)\xi^2\right] r dr \tag{IX.1}$$

with $W_0 = \frac{2\pi^2 R_0 B_0^2}{\mu_0 a^2}$. When $m = n = 1$, if there exists an $\iota = 1$ surface in the plasma, then a trial function can be constructed such that $\delta\hat{W}_2 \to 0$ by letting $\xi'(r) = 0$ except very close to the $\iota = 1$ surface. In this case, the next-order contribution to δW is fourth-order in ε, with

$$\delta\hat{W}_4 = \frac{\delta W_4}{\varepsilon^4 W_0} = \xi_0^2 \int_0^{r_1} \left[r\beta' + \frac{r^2}{R_0^2}(1-\iota)(3+\iota)\right] \tag{IX.2}$$

Where $\iota(r_1) = 1$ and $\beta'(r) = \frac{p'(r)}{B_0^2/2\mu_0}$.

IX.III.ii Comparison to results from eigenvalue solver

The analytic results for the $m = 1$ internal kink mode were then compared to numerical results from the eigenvalue solver; ι_0 was varied from 0.8 to 0.125 for a series of equilibria with

$$\iota(\rho) = \iota_0 - 0.1\rho^2 \tag{IX.3}$$

Figure 9 demonstrates the poloidal dependence of the least stable eigenfunction for $\iota_0 = 1.1$ — the eigenfunction clearly has a poloidal mode number $m = 1$, aligning with the expectation that the $m = 1$ kink mode should be the only unstable fixed-boundary mode for the device.

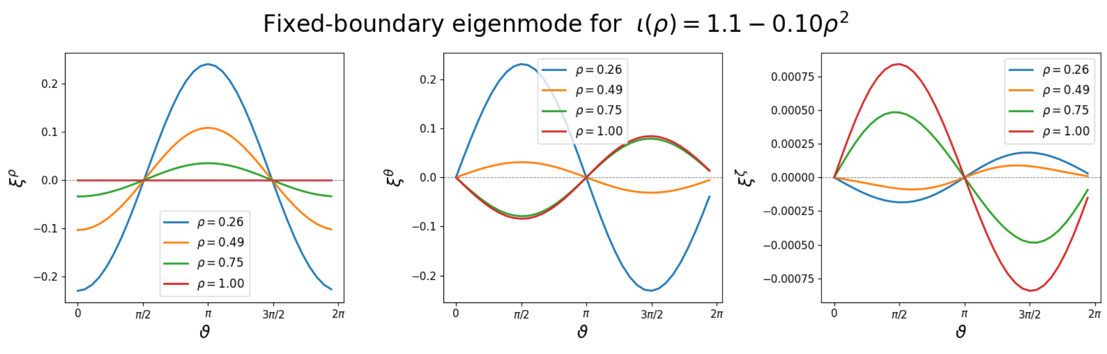

Figure 9: Poloidal dependence of the least stable eigenfunction for $\iota(\rho) = 1.1 - 0.1\rho^2$. The structure of the eigenmode closely resembles an $m = 1$ internal kink mode.

By computing the Chebyshev-Fourier spectral fit to the perturbation and then taking the sum over radial modes, we can obtain a more quantitative estimate of the mode number of the perturbation. As seen in Figure 10, the $m = 1$ mode is completely dominant throughout the entire unstable domain, as expected from IX.III.i.

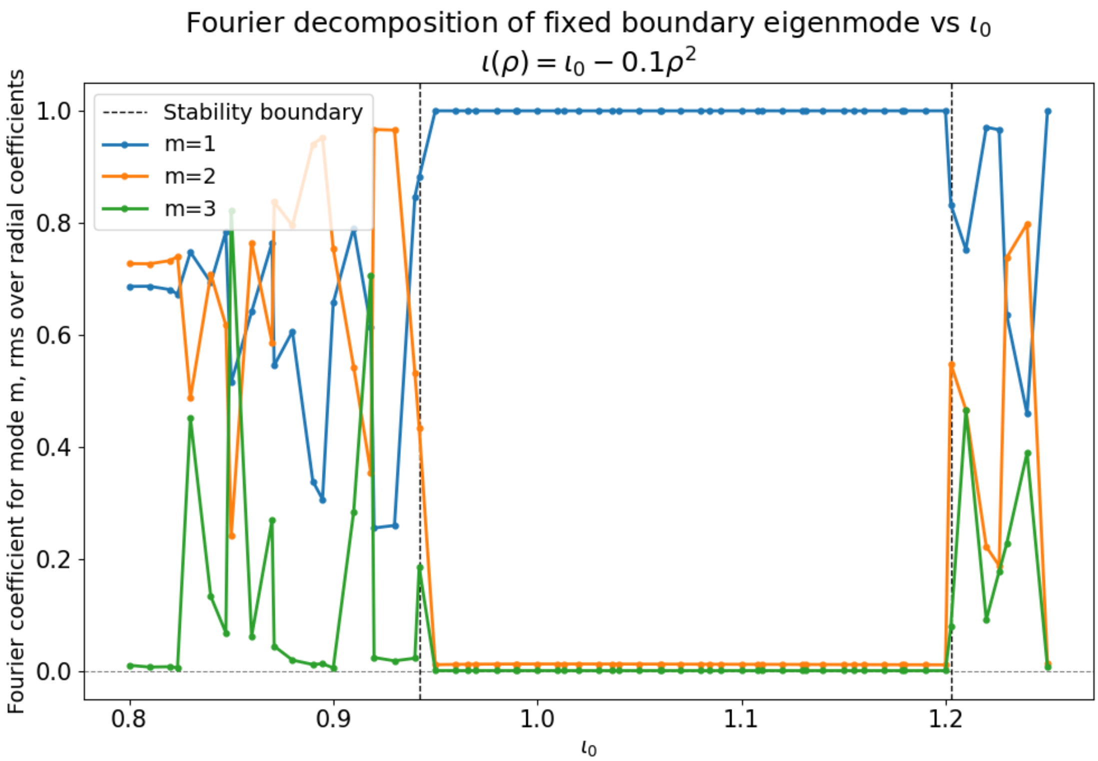

Figure 10: Poloidal mode number of the fixed boundary eigenmode for the ι profiles (IX.3). The $m = 1$ Fourier harmonic is dominant for all unstable modes.

Figure 11 shows the stability eigenvalue as a function of the ι_0 on axis. The eigenmode becomes unstable for $\iota_0 > 0.94$, and becomes stable for $\iota_0 \geq 1.2$; since there is no longer a surface where $\iota_0 - 1$ is small. The eigenmode is least stable for $\iota_0 = 1.08$; this can be expected — for $\iota_0 = 1.08$, the $\iota = 1$ surface is within the plasma volume, so that the stabilizing contribution of (IX.1) can be minimized. However, the $\iota = 1$ surface is still close to the edge of the plasma, so that the destabilizing integral (IX.2) is taken over as much of the radial domain as possible.

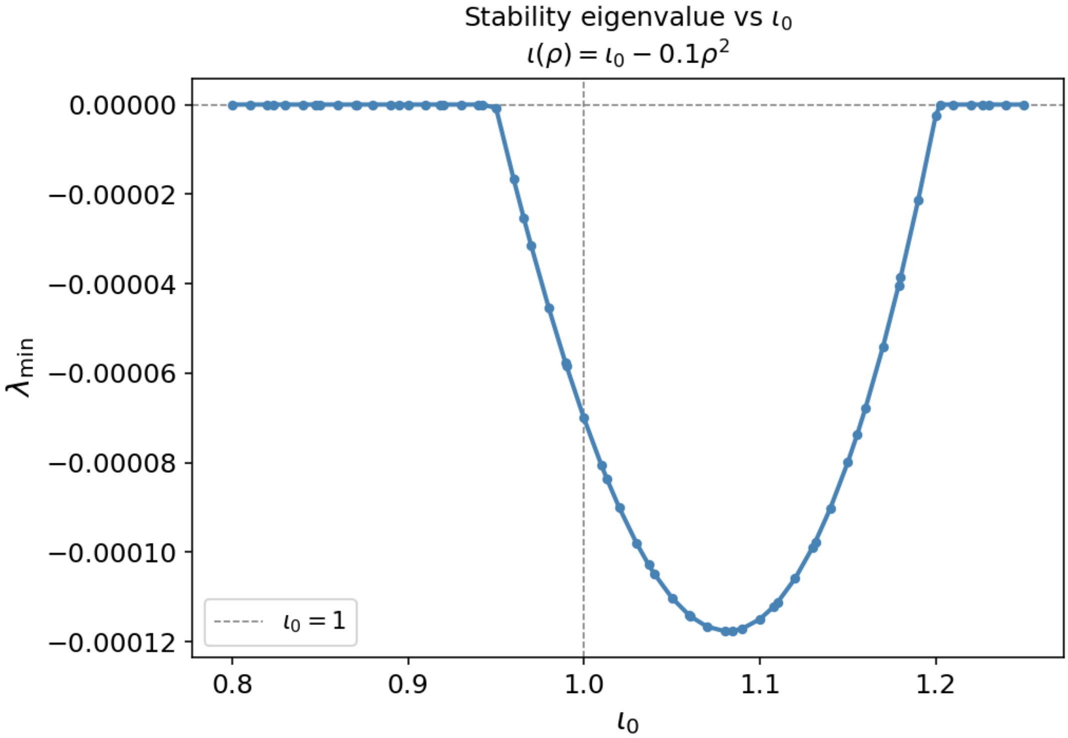

Figure 11: Stability eigenvalue for fixed-boundary modes for the ι profiles (IX.3). The equilibria are unstable for $\iota_0 \in (0.94, 1.2)$, with the most unstable equilibrium at $\iota_0 = 1.08$.

The instability of the eigenvalue outside the range $\iota_0 \in [1.0, 1.1]$ is likely a product of the non-negligible $\varepsilon = \frac{a}{R_0} = \frac{1}{10}$; near the axis or edge (respectively), $|\iota - 1|$ may be sufficiently small that higher-order terms in δW can dominate over the purely stabilizing δW_2 and drive instability.

IX.IV External kink modes

IX.IV.i Analytic results

When the fixed-boundary requirement is relaxed, the straight tokamak is significantly less stable. In the case of a conducting wall at infinity, the perturbed potential energy is given by

$$\delta \hat{W}_2 = \int_0^a \left(\frac{n}{m} - \iota\right)^2 \left[r^2 \xi'^2 + (m^2 - 1)\xi^2\right] r \, dr$$
$$+ \left(\frac{n}{m} - \iota_a\right) \left[\left(\frac{n}{m} + \iota_a\right) + m \left(\frac{n}{m} - \iota_a\right)\right]$$

where ι_a is the rotational transform at the edge.

External $m = 1$ kink mode The minimizing trial function for the $m = 1$ mode is constant as a function of r, such that the purely stabilizing internal energy δW_F vanishes. In the case of the $m = 1$ mode, the perturbed energy is given by

$$\delta \hat{W}_2 = 2a^2 \xi_a^2 \iota_a n \left(\frac{n}{\iota_a} - 1 \right)$$

Because this eigenvalue minimizes δW_2, the eigenfunction $\xi(\rho) = \xi_0$ is also a true eigenfunction. The $n = m = 1$ external kink mode is therefore unstable for $\iota_a > 1$. Observe also that the stability of the $m = 1$ external kink mode is independent of the rotational transform profile — the eigenvalue depends only on ι_a, and therefore the total current through the plasma, but not the profile as a function of ρ.

External $m = 2$ kink mode Unlike the $m = 1$ mode, the stability boundary of the $m = 2$ mode depends sensitively on the particular rotational transform profile used. For instance, a constant ι profile is always unstable, regardless of the value of ι.

On the other hand, if the current density vanishes at the edge, but the edge gradient of the current density is linear, bands of stability are possible. For instance, the following series of rotational transform profiles is considered:

$$\iota(\rho) = \frac{1}{\alpha^2 \rho^2} \left[\alpha \rho^2 + (1 - \alpha) \ln(1 - \alpha \rho^2) \right] \tag{IX.4a}$$

$$\iota(1) = \frac{1}{\alpha^2} \left[\alpha + (1 - \alpha) \ln(1 - \alpha) \right] \tag{IX.4b}$$

Where $\alpha \in (-\infty, 1)$ is a free parameter. For this series of profiles, bands of stability are possible for $\iota_a < \frac{1}{2}$, but $\iota_a > \frac{1}{2}$ is always unstable to the $m = 2$ kink mode (Figure 12). Furthermore, the $m = 3$ and $m = 4$ modes have significantly smaller growth rates than the $m = 2$ mode.

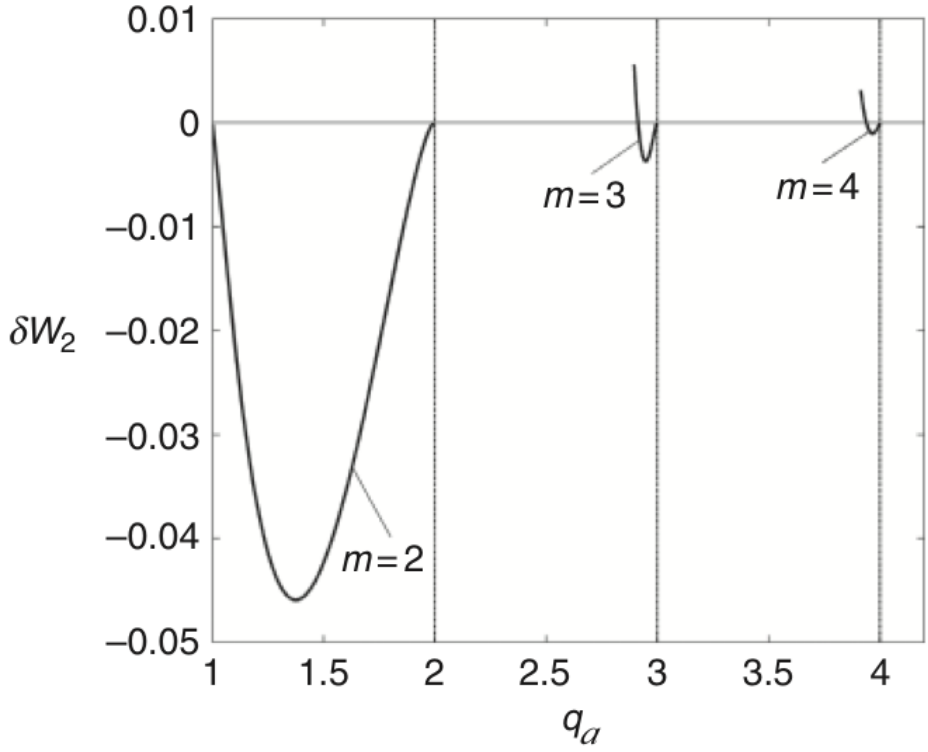

Figure 12: Stability of the straight tokamak to the $n = 1$ mode for varying poloidal mode number m and the profiles given by (IX.4). From Freidberg [7].

IX.IV.ii Results from eigenvalue solver

Quadratic ι profile The equilibria analyzed for stability to internal modes in IX.III.ii were reanalyzed for stability to external modes. When the plasma boundary was allowed to change, the equilibria were unstable for all $\iota_0 \in [0.8, 1.25]$, with the eigenvalue decreasing linearly with ι_0. The typical magnitude of $\lambda_{\min}$ was also O(ε^{-2})=O(100) times larger than the eigenvalues seen in IX.III.ii, aligning with the expectation that external modes should have $\delta W \sim O(\varepsilon^2)$, whereas for internal modes, the first non-vanishing unstable contribution to δW is $O(\varepsilon^4)$.

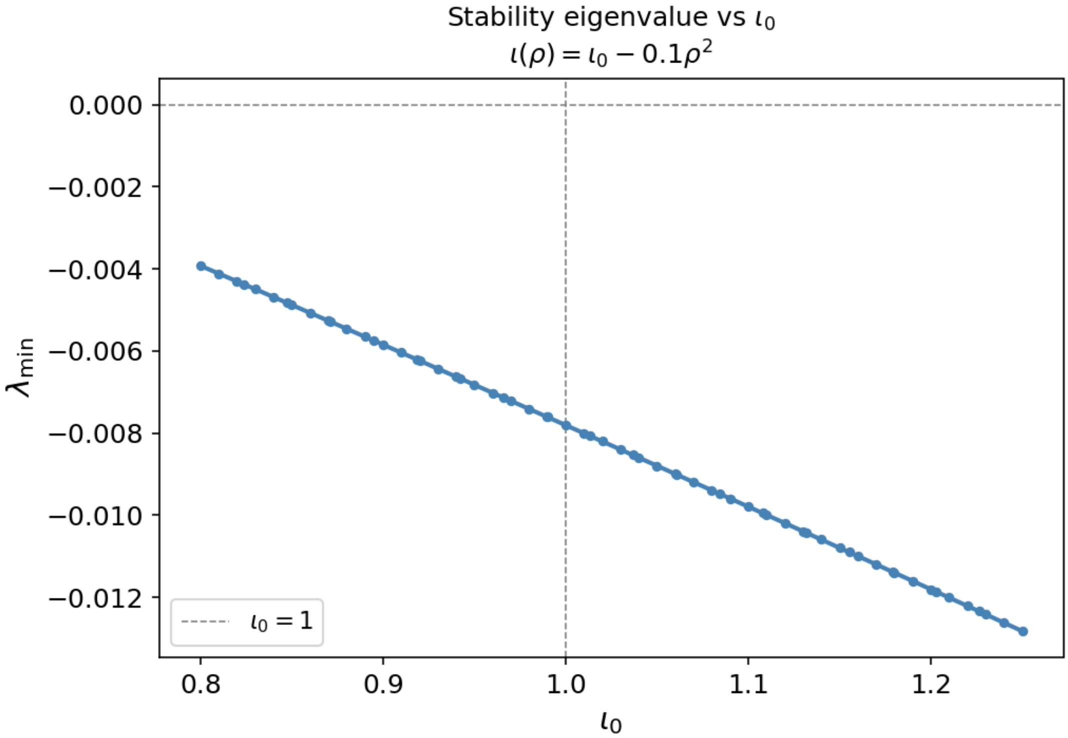

Figure 13: Stability eigenvalue for free-boundary modes for the ι profiles (IX.3). The equilibria are unstable for all $\iota_0 \in [0.8, 1.25]$, with the eigenvalue decreasing linearly in ι_0.

Furthermore, the dominant mode number has become the $m = 2$ mode (Figure 14), which for fixed-boundary modes is always stable. However, for $\iota_0 > 1.1$, i.e. $\iota_a > 1$, the $m = 1$ Fourier mode begins to grow with ι_0. This aligns with the expectation that the external $m = 1$ kink mode should become unstable for $\iota_a > 1$.

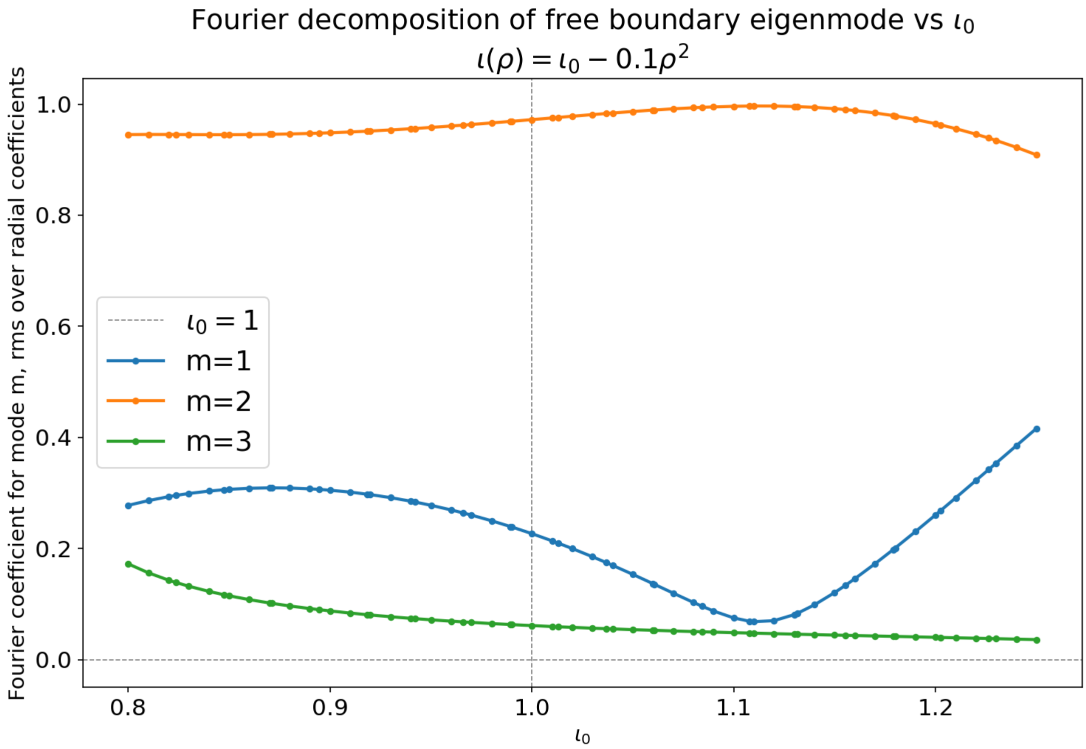

Figure 14: Poloidal mode number of the free boundary eigenmode for the ι profiles (IX.3). The $m = 2$ mode is generally dominant.

Linear current density edge gradient The next series of equilibria considered are the profiles defined by (IX.4). α was varied from -2 to 0.995, equivalent to varying ι_a between 0.32 and 0.98.

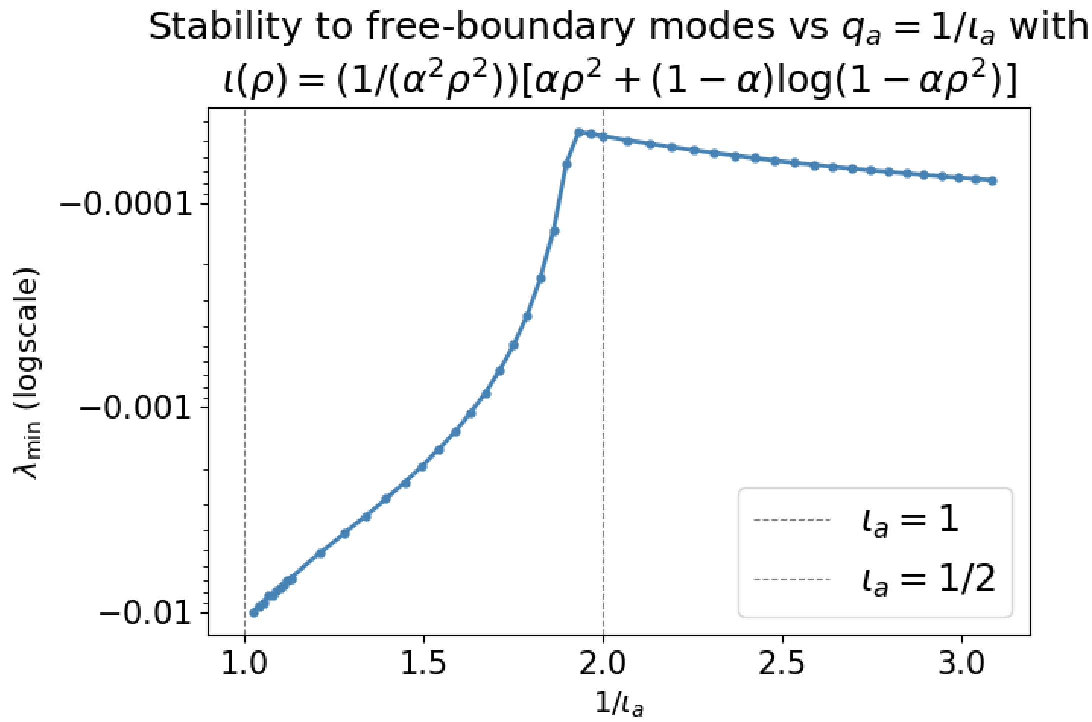

Figure 15: Stability eigenvalue for free-boundary modes for the ι profiles (IX.4). A logarithmic scale is used to show the separation of scales between the internal and external kink modes, and the edge safety factor $q_a = \frac{1}{\iota_a}$ is used for consistency with Figure 12.

As observed in Figure 15, for $\iota_a > \frac{1}{2}$, the equilibria are extremely unstable, with $\lambda_{\min}$ as negative as -0.01.

However, as $q_a = \frac{1}{\iota_a}$ approaches 2, the equilibria become dramatically more stable, aligning with the expectation from Figure 12 that the plasma should be stable to external kink modes for $q_a > 2$, other than narrow bands just below $q_a = 3$ and $q_a = 4$. However, since $\iota_0 = 1$ for all equilibria considered in this series, all equilibria are still unstable to internal kink modes. Indeed, we see in Figure 16 that the radial structure of the mode aligns closely with that of an internal kink mode; $|\xi|$ peaks near the axis (the $\iota = 1$ surface), and is nearly 0 at the boundary ($\sim 10^{-5}$).

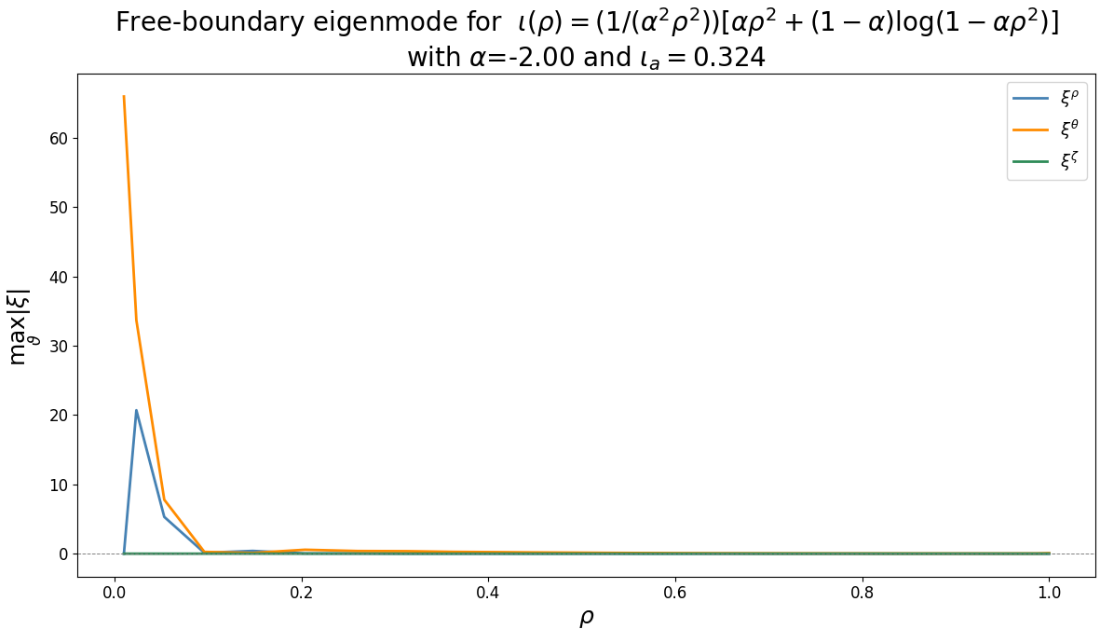

Figure 16: Radial structure of the least stable eigenmode for the profile (IX.4) with $\alpha = -2.00$. Although the Dirichlet boundary condition is not enforced, the $m = 1$ internal kink mode is still the dominant eigenmode.

This analysis is further confirmed by a Fourier decomposition of ξ^ρ for varying q_a (Figure 17). For all $q_a \in [2, 3]$, the $m = 1$ mode is dominant, as would be expected for the internal kink mode in a straight tokamak.

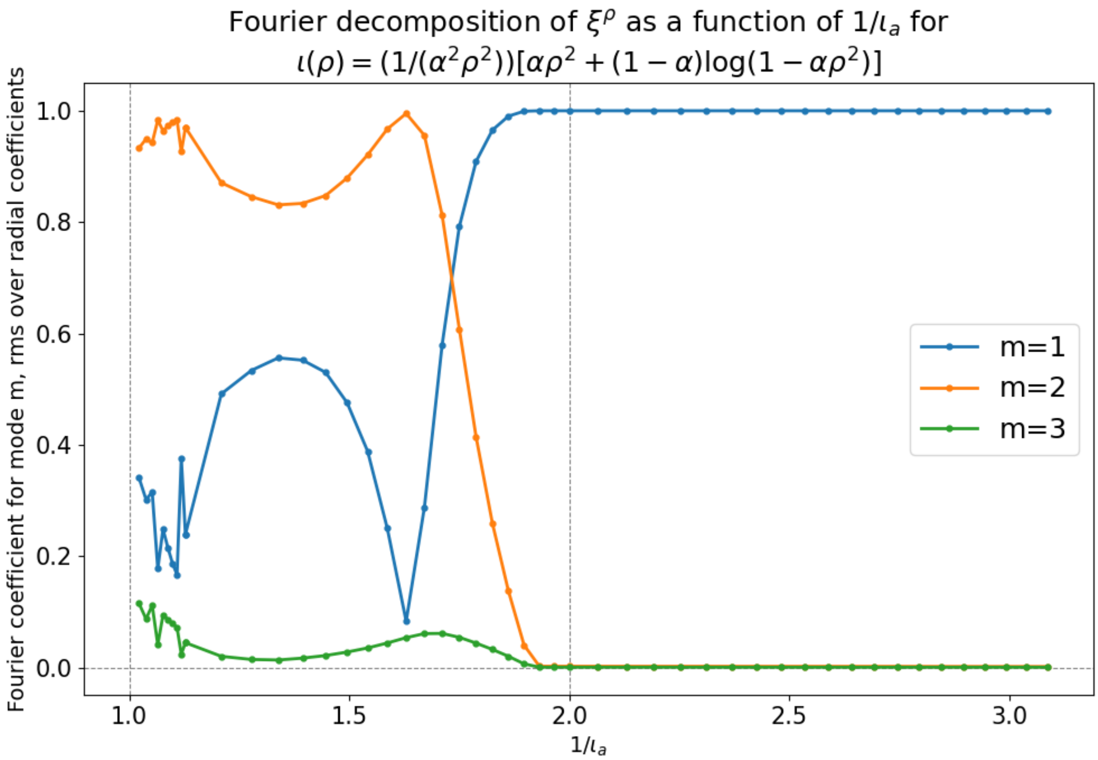

Figure 17: Fourier decomposition of the least stable eigenfunction for the profiles described by (IX.4). The $m = 2$ mode is dominant for $q_a = \frac{1}{\iota_a} \lesssim 1.6$, but for $q_a > 2$, the mode is purely $m = 1$, corresponding to an internal kink mode.

Note that we do not see the $m = 3$ external kink mode even as $q_a \to 3$; it appears the $m = 1$ internal kink mode dominates over the $m = 3$ external kink mode, likely a result of the non-neglible inverse aspect ratio $\varepsilon = \frac{1}{10}$ and the fact that the external $m = 3$ kink mode is quite weak in this series of profiles (as seen in Figure 12).

On the left hand side of Figure 15, we can see that the plasma continues to become increasingly unstable as $q_a \to 1$. While at first glance, this appears to contradict Figure 12, we can see in Figure 17 that the $m = 1$ mode becomes more dominant in this region. Since the finite toroidicity couples poloidal modes, and the equilibria considered here are always unstable to external kink modes for all $q_a < 2$, it is not unexpected that the equilibria become more unstable as ι_a approaches the instability regime of the $m = 1$ mode. Figure 18 illustrates a mixed $m = 1/m = 2$ eigenmode near $\frac{1}{\iota_a} = 1$. In comparison, Figure 19 depicts a nearly pure $m = 2$ mode at $\frac{1}{\iota_a} \sim 1.6$, where the $m = 1$ external kink mode is stable.

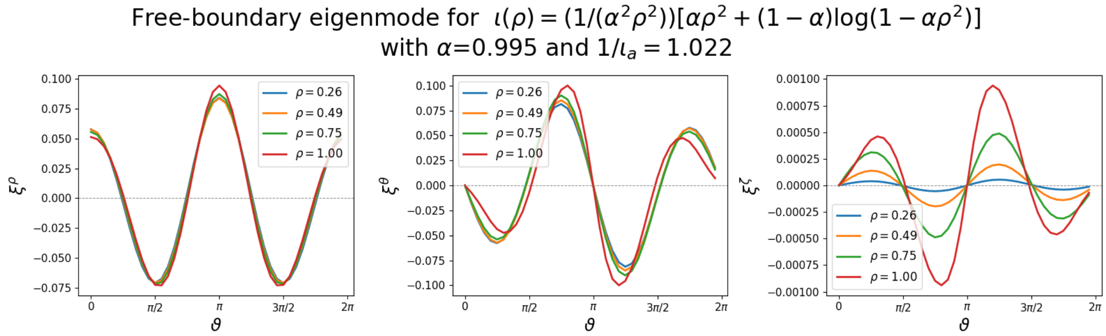

Figure 18: Mixed $m = 1$ and $m = 2$ mode for $\iota_a \to 1$

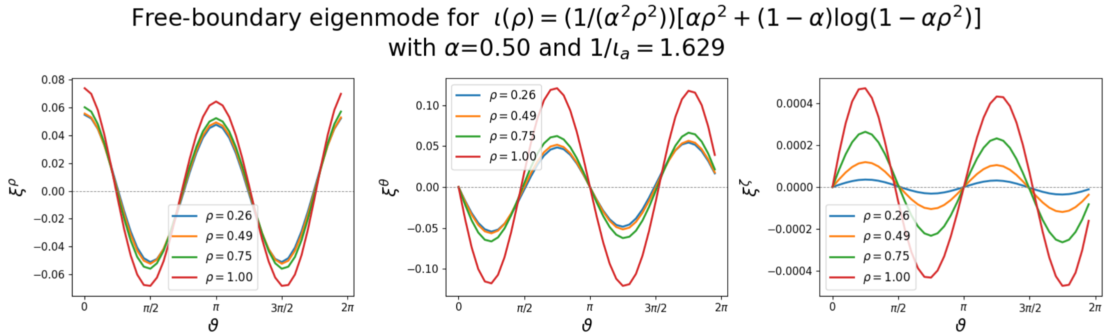

Figure 19: Pure $m = 2$ external kink mode, for $1/\iota_a \sim 1.6$.

X Conclusion

This work contains an overview of the principles and numerical methods used in ideal magnetohydrodynamics, particularly as they apply to magnetic confinement fusion and MHD stability in particular. Two simple exercises were performed to illustrate the numerical methods used in ideal MHD and provide an intuition for the stability properties of 1D equilibria.

AGNI, the open-source fixed-boundary MHD stability eigenvalue solver currently under development by Rahul Gaur, has been extended to evaluate stability to external modes. The vacuum energy is computed precisely, and the evaluation of the singular integrals used to solve the exterior Neumann problem is accurate to >10th order in the number of grid points. The accuracy of the solution to the exterior Neumann problem and the integration of the vacuum energy is demonstrated through two simple test cases. Finally, both the fixed-boundary and the free-boundary versions of AGNI have been validated through comparison with analytic results from Chapter 11 of Freidberg [7]. This work enables stability calculations directly in the DESC suite and lays the groundwork for automatically differentiable free-boundary stability calculations to be used during equilibrium optimization.

XI Future work

Validation of stability to external modes in non-axisymmetric geometries The immediate next step to pursue is to validate the free-boundary stability computations in more complex geometries. As discussed, validation of 3D stability calculations is challenging, since analytic results are very difficult to obtain and there are no open-source MHD stability codes available for comparison. However, Merkel et al. [25] describes free-boundary stability results for a sequence of Wendelstein 7-AS (W7-AS) stellarator equilibria with varying imposed vertical field B_Z, and obtains a stability boundary at $\frac{B_Z}{B_0} \sim 0.014$. A natural next step would be to recreate the results described in that work, in order to benchmark the free-boundary version of AGNI against existing MHD stability solvers.

MHD stability optimization As discussed in VI.II.i, currently the only way to optimize the stability eigenvalues of an equilibrium is by running TERPSICHORE or CAS3D many times each optimization step, and then using finite differences to compute the gradients of the stability eigenvalue with respect to the optimizable parameters. Once the extension of the eigenvalue solver to external modes has been validated, a useful next step would be to use the least stable eigenvalue as an objective function during equilibrium optimization. Nonlinear MHD simulations in M3D-C1 [32] could be then be used to validate the success of the optimization in improving the stability of the equilibrium, as in the design of Thea Energy's Helios equilibrium [33].

Anisotropic plasmas Another interesting path to explore further is the effect of anisotropic pressure (i.e. $p_\parallel \neq p_\perp$) on stability. DESC already has the capability to consider anisotropic plasmas at equilibrium. Cooper et al. [34] extends the TERPSICHORE code to consider the effects of pressure anisotropy in two limiting cases meant to represent the impact of energetic particles (created by sources of auxiliary heating such as neutral beam injection) on stability. The more restrictive limit uses the potential energy derived in Kruskal et al. [35] while ignoring the stabilizing kinetic integral. The second, less restrictive, limit corresponds to the regime in which energetic particles in the plasma drift more rapidly than the growth rate of MHD instabilities. Both limits are described as an energy principle. A potential area for future work would be to develop a theoretical model for interactions between energetic particles and MHD instabilities, and implement this as an energy principle in AGNI.

Lagrangian approach and extension to nonlinear MHD stability Kolemen [36] has suggested a considerably simpler method for MHD stability calculations may be possible through JAX automatic differentiation. Since the force operator $\mathbf{f} \equiv \mathbf{J} \times \mathbf{B} - \nabla p$ is directly related to δW, differentiating $\mathbf{f}$ automatically using JAX could yield the same results as the variational approach more efficiently. The eigenvalue equation (VI.3a) could then be solved directly.

This method could be used to extend stability analysis to nonlinear stability as well. Experimental results from Wendelstein-7X (W-7X) have indicated that stellarators can operate above the β limits set by linear stability criteria like Mercier's criterion, though the mechanism is not yet understood. Nonideal MHD simulation tools such as M3D-C1 and Jorek have recently been extended to stellarator geometries, confirming that stable operation above linear β limits is possible, but requiring computationally demanding simulations that are impossible to optimize for [37] [38]. Chu et al. [39] analytically derives the nonlinear saturation condition of infinite-n ballooning modes in stellarators, reproducing nonlinear behavior observed in W-7X. However, Zhou et al. [37] and Ramasamy et al. [38] have motivated nonlinear analysis of low-n instabilities in stellarators, on which no analytic progress has yet been made.

Using the Lagrangian approach, higher-order derivatives of the force balance equation — easily obtainable through JAX automatic differentiation — could be used to predict observable nonlinear phenomena such as small-amplitude benign saturation of instabilities and energy release.

Statement on the use of generative AI This work acknowledges the use of generative artificial intelligence for assistance in basic programming applications.

References

[1] International Atomic Energy Agency. *Fundamentals of Magnetic Fusion Technology*. International Atomic Energy Agency, Vienna, 2023.

[2] International Energy Agency. Global energy review 2025. Technical report, IEA, Paris, March 2025. Licence: CC BY 4.0.

[3] U.S. Department of Energy. Doe explains...tokamaks, n.d. Accessed: April 23, 2026.

[4] R. J. Goldston and P. H. Rutherford. *Introduction to Plasma Physics*. CRC Press, 1st edition, 1995.

[5] Lise-Marie Imbert-Gerard, Elizabeth J. Paul, and Adelle M. Wright. An introduction to stellarators: From magnetic fields to symmetries and optimization, 2020.

[6] Robert Goldston. AST309/MAE309/PHY309/ENE309: The Science of Fission and Fusion Energy. Course Lectures, Princeton University, 2024. Spring Semester.

[7] Jeffrey P. freidberg. *Ideal MHD*. Cambridge University Press, 2014.

[8] Matthew Kunz. AST552: General Plasma Physics II. Course Lectures, Princeton University, 2026. Spring Semester.

[9] John P. Boyd. *Chebyshev and Fourier Spectral Methods*. Dover Publications, Mineola, NY, 2nd edition, 2001.

[10] Allen H. Boozer. Transport and isomorphic equilibria. *The Physics of Fluids*, 26(2):496–499, 02 1983.

[11] D.W. Dudt, R. Conlin, D. Panici, and E. Kolemen. The desc stellarator code suite part 3: Quasi-symmetry optimization. *Journal of Plasma Physics*, 89(2), April 2023.

[12] PlasmaControl. General, 2024. DESC v0.16.0 documentation.

[13] M. D. Kruskal and R. M. Kulsrud. Equilibrium of a Magnetically Confined Plasma in a Toroid. *The Physics of Fluids*, 1(4):265–274, July 1958. Publisher: American Institute of Physics.

[14] James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas,

Skye Wanderman-Milne, and Qiao Zhang. JAX: composable transformations of Python+NumPy programs.

[15] David V. Anderson, W. Anthony Cooper, Ralf Gruber, Silvio Merazzi, and Ulrich Schwenn. *TERPSICHORE: A Three-Dimensional Ideal Magnetohydrodynamic Stability Program*, pages 159–174. Springer US, Boston, MA, 1990.

[16] Carolin Schwab. Ideal magnetohydrodynamics: Global mode analysis of three-dimensional plasma configurations. *Physics of Fluids B: Plasma Physics*, 5(9):3195–3206, 09 1993.

[17] Dario Panici, Rory Conlin, Daniel W. Dudt, Kaya Unalmis, and Egemen Kolemen. The desc stellarator code suite part i: Quick and accurate equilibria computations, 2023.

[18] S. P. Hirshman and J. C. Whitson. Steepest descent moment method for three dimensional magnetohydrodynamic equilibria. *Phys. Fluids*, 26:3553–3568, 1983.

[19] Rahul Gaur. DESC: Stellarator optimization suite, 2026. Branch implementing MHD stability to internal modes, manuscript in preparation.

[20] Claudio Canuto, Mohammed Hussaini, Alfio Quarteroni, and Thomas Zang. *Spectral Methods: Fundamentals in Single Domains*, volume 23. 01 2006.

[21] Lloyd N. Trefethen. *Spectral Methods in MATLAB*. Software, Environments, and Tools. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2000.

[22] SciPy Contributors. `scipy.linalg.eigh` — SciPy v1.17.0 manual, 2025. Accessed: 2026-04-26.

[23] SciPy Contributors. `scipy.sparse.linalg.eigsh` — SciPy v1.17.0 manual, 2025. Accessed: 2026-04-26.

[24] Rahul Gaur. A novel 3D MHD stability solver to investigate millions of generalized omnigenous equilibria in DESC. Poster presented at the International Stellarator-Heliotron Workshop (ISHW), 2026. ISHW 2026.

[25] P. Merkel, C. Nuehrenberg, and W.A. Cooper. Free-boundary ideal mhd modes in w7-as. page p. 51–58., Sep 1996.

[26] Peter Merkel. An integral equation technique for the exterior and interior neumann problem in toroidal regions. *Journal of Computational Physics*, 66(1):83–98, 1986.

[27] Rory Conlin, Jonathan Schilling, Daniel W. Dudt, Dario Panici, Rogerio Jorge, and Egemen Kolemen. High order free boundary mhd equilibria in desc, 2024.

[28] Dhairya Malhotra, Antoine J Cerfon, Michael O'Neil, and Evan Toler. Efficient high-order singular quadrature schemes in magnetic fusion. *Plasma Physics and Controlled Fusion*, 62(2):024004, December 2019.

[29] Dhairya Malhotra, Antoine Cerfon, Lise-Marie Imbert-Gérard, and Michael O'Neil. Taylor states in stellarators: A fast high-order boundary integral solver. *Journal of Computational Physics*, 397:108791, November 2019.

[30] Kaya Unalmis. New high-order accurate free surface stellarator equilibria optimization and boundary integral methods in DESC. Senior thesis, Princeton University, Princeton, NJ, April 2025.

[31] M. C. Zarnstorff, L. A. Berry, A. Brooks, E. Fredrickson, G.-Y. Fu, S. Hirshman, S. Hudson, L.-P. Ku, E. Lazarus, D. Mikkelsen, D. Monticello, G. H. Neilson, N. Pomphrey, A. Reiman, D. Spong, D. Strickler, A. Boozer, W. A. Cooper, R. Goldston, R. Hatcher, M. Isaev, C. Kessel, J. Lewandowski, J. F. Lyon, P. Merkel, H. Mynick, B. E. Nelson, C. Nuehrenberg, M. Redi, W. Reiersen, P. Rutherford, R. Sanchez, J. Schmidt, and R. B. White. Physics of the compact advanced stellarator NCSX. *Plasma Physics and Controlled Fusion*, 43(12A):A237, November 2001.

[32] Nate Ferraro, Stephen Jardin, Brendan Lyons, Seegyoung Seol, Jin Chen, Yao Zhou, Chang Liu, Andreas Kleiner, and Joshua Breslau. M3d-c1. [Computer Software] `https://doi.org/10.11578/dc.20180627.9`, jun 2018.

[33] C. P. S. Swanson, S. T. A. Kumar, D. W. Dudt, E. R. Flom, W. B. Kalb, T. G. Kruger, M. F. Martin, J. R. Olatunji, S. Pasmann, L. Z. Tang, J. von der Linden, J. Wasserman, M. Avida, A. S. Basurto, M. Dickerson, N. de Boer, M. J. Donovan, A. H. Doudna Cate, D. Fort, W. Harris, U. Khera, A. Koen, J. A. Labbate, N. Maitra, A. Ottaviano, R. K. Parmar, E. J. Paul, B. Reydel, A. van Riel, P. K. Romano, M. Savastianov, S. Saxena, S. Seethalla, S. Srinivasan, R. H. Wu, D. Nash, J. Priebe, M. Slepchenkov, S. Walsh, B. Berzin,

D. A. Gates, and the Thea Energy team. Overview of the helios design: A practical planar coil stellarator fusion power plant, 2026.

[34] W. A. Cooper, J. P. Graves, T. M. Tran, R. Gruber, T. Yamaguchi, Y. Narushima, S. Okamura, S. Sakakibara, C. Suzuki, K. Y. Watanabe, H. Yamada, and K. Yamazaki. Stability properties of anisotropic pressure stellarator plasmas with fluid and noninteractive energetic particles. *Fusion Science and Technology*, 50(2):245–257, 2006.

[35] M. D. Kruskal and C. R. Oberman. On the stability of plasma in static equilibrium. *The Physics of Fluids*, 1(4):275–280, 07 1958.

[36] Egemen Kolemen. Lagrangian ideal mhd stability using spectral decomposition in desc. Manuscript, May 2024.

[37] Yao Zhou, K. Aleynikova, Chang Liu, and N.M. Ferraro. Benign saturation of ideal ballooning instability in a high-performance stellarator. *Physical Review Letters*, 133(13), 2024.

[38] Rohan Ramasamy, Haowei Zhang, Joachim Geiger, Carolin Nührenberg, Håkan M. Smith, Karl Lackner, and Valentin and Igochine. A comparison of low-n mercier unstable wendelstein stellarators and quasi-interchange modes in tokamaks. *Journal of Plasma Physics*, 91(4), August 2025.

[39] X. Chu, S. C. Cowley, N. Ferraro, Y. Zhou, and F. I. Parra. Nonlinear saturation of ballooning modes in stellarators, 2026.

A Curvilinear coordinate systems

This appendix contains some useful definitions and formulas for general curvilinear (u^1, u^2, u^3) coordinate systems that are frequently quoted throughout this thesis. Throughout, repeated indices indicate summation.

Covariant basis vectors

$$\mathbf{e}_i \equiv \frac{\partial \mathbf{x}}{\partial u^i} \tag{A.1}$$

Contravariant basis vectors

$$\mathbf{e}^i \equiv \nabla u^i \tag{A.2}$$

Reciprocal vectors

$$\mathbf{e}^i \cdot \mathbf{e}_j = \delta^i_j \tag{A.3}$$

Covariant components of a vector

$$\mathbf{D} = D_i \mathbf{e}^i \quad \text{with} \quad D_i \equiv \mathbf{D} \cdot \mathbf{e}_i \tag{A.4}$$

Contravariant components of a vector

$$\mathbf{D} = D^i \mathbf{e}_i \quad \text{with} \quad D^i \equiv \mathbf{D} \cdot \mathbf{e}^i \tag{A.5}$$

Covariant metric tensor

$$g_{ij} \equiv \mathbf{e}_i \cdot \mathbf{e}_j \tag{A.6}$$

Contravariant metric tensor

$$g^{ij} \equiv \mathbf{e}^i \cdot \mathbf{e}^j \tag{A.7}$$

Jacobian

$$\begin{aligned} &J \equiv \frac{\partial(x, y, z)}{\partial(u^1, u^2, u^3)} = \mathbf{e}_1 \cdot \mathbf{e}_2 \times \mathbf{e}_3 = \frac{1}{\mathbf{e}^1 \cdot \mathbf{e}^2 \times \mathbf{e}^3} = \sqrt{g} \\ &\text{with} \quad g \equiv \det[g_{ij}] = \frac{1}{\det[g^{ij}]} \end{aligned} \tag{A.8}$$

Divergence

$$\nabla \cdot \mathbf{D} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^i} \left(\sqrt{g} D^i \right) \tag{A.9}$$

Curl

$$\nabla \times \mathbf{D} = \frac{\varepsilon^{ijk}}{\sqrt{g}} \frac{\partial D_j}{\partial u^i} \mathbf{e}_k \tag{A.10}$$